

Downstream Effects of Biofuel Policy Shocks

Jungkeon Jo, Xiaoli Etienne, and Michael K. Adjemian*

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Abstract

We study how U.S. biofuel policy shocks transmit through agri-food supply chains to downstream markets, including consumer prices, food prices, and food expenditures. Drawing on the institutional structure of U.S. biofuel policy and high-frequency price movements around regulatory announcements from the EPA, CARB, and Reuters, we construct a novel biofuel policy news series and use it as an external instrument in a structural VAR framework to estimate the dynamic causal effects of biofuel policy shocks. A stringent biofuel policy shock leads to significant inflation across both upstream and downstream markets. Biofuel, oil, and field crop prices rise substantially; in the downstream market, consumer food prices and the headline CPI increase, while household food spending declines. Specifically, fats and oils prices rise more sharply than other food categories given their direct role in biofuel feedstock competition, while producer food prices respond more strongly than their CPI counterparts, indicating that upstream cost pressures are partially absorbed before reaching consumers. Retail energy prices and consumer inflation expectations also rise following the shock. The inflationary burden is not evenly distributed—lower-middle income households face larger price increases than upper-middle-income households, making the shock regressive. These findings reveal a fundamental trade-off in biofuel policy. While higher commodity prices support agricultural producer revenues and strengthen U.S. energy independence, consumers finance these gains through higher retail prices for both fuel and food, with lower-income households bearing a disproportionate burden.

Keywords: Biofuel policy, Renewable Fuel Standard, high-frequency identification, external instrument SVAR, food price inflation, price transmission, consumer prices, renewable diesel

JEL classification: Q42, Q18, E31, C32, Q11, Q48

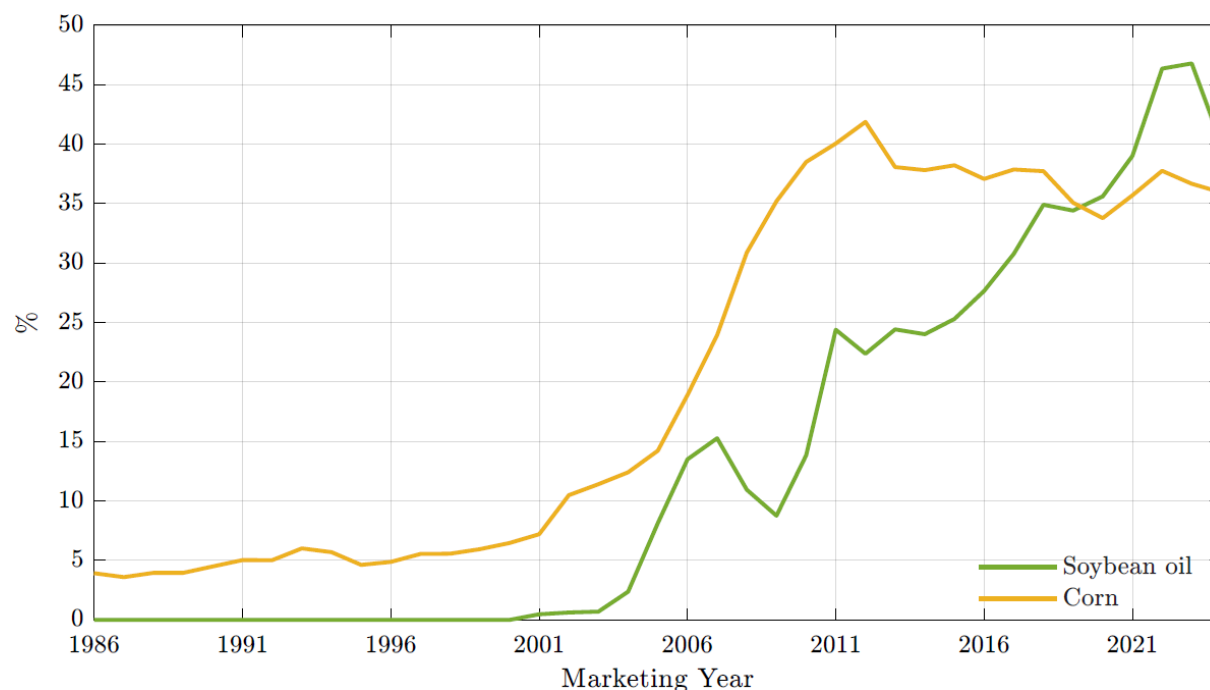
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1. Introduction

Energy and food prices are among the most politically sensitive components of inflation, directly affecting consumers through gasoline and grocery expenditures. One federal policy in the United States linking the two is the Renewable Fuel Standard (RFS), which mandates minimum volumes of renewable fuels blended into the U.S. transportation fuel supply (EPA, 2025a). By expanding demand for agricultural commodities, the mid-2000s biofuel boom supported farm incomes and rural development through higher commodity prices and investment (Rajagopal and Zilberman, 2007; Moschini and Lapan, 2012).

Today, biofuels represent a primary outlet for U.S. agricultural commodities, with the scale of this dependence growing rapidly over time. As illustrated in Figure 1, over the last decade, 35-40 percent of U.S. corn production has been allocated to ethanol manufacturing. At the same time, the biodiesel and renewable diesel sectors have absorbed a rapidly growing share of oilseed output, with biofuel use of soybean oil surging from less than 1 percent in 2001/02 to over 40 percent by 2024/25 (Avileis, 2025).

Figure 1. Shares of U.S. corn for ethanol and soybean oil for biofuel production



Source: Table 5. 'Corn supply, disappearance (million bushels), and share of total corn used for fuel alcohol (percent)' & Table 6. 'Soybean oil supply, disappearance (million pounds), and share used for biofuel (percent)' in USDA Bioenergy Statistics.

This structural transformation has important implications for downstream food and energy markets. Under the RFS, obligated parties, primarily refiners and importers, must acquire and retire sufficient Renewable Identification Numbers (RINs) annually to comply with mandated blending volumes. As Lade and Bushnell (2019) show, RFS mandates impose implicit taxes on fossil fuels and implicit subsidies for biofuels, with both adjusting endogenously in response to policy stringency. Policy actions that tighten enforcement or increase compliance costs can affect fuel prices and alter demand for agricultural feedstocks, creating potential spillovers to food markets (de Gorter and Just, 2009; Lade and Bushnell, 2019).

These linkages motivate a central question: to what extent do policy-driven shifts in biofuel demand propagate and feed into broader inflationary pressures? This question is directly relevant to the longstanding “food versus fuel” debate, or whether diverting agricultural commodities toward energy production raises food prices and threatens food security (von Braun et al, 2008; Hochman et al., 2011; Zilberman et al., 2013). Proponents of biofuel mandates argue that price effects are modest and temporary, as supply responses and productivity gains offset demand pressures (de Gorter and Just, 2010). Critics counter that even modest increases in commodity prices translate into meaningful food price inflation for vulnerable consumers, with disproportionate burdens on low-income households who spend larger shares of their budgets on food and energy (Wright, 2011; Wright, 2014).

Previous research has documented significant effects of biofuel policies on upstream agricultural commodity markets. During the initial expansion of corn ethanol production following the 2005 RFS, biofuel demand was identified as a key driver of corn price movements (Carter et al., 2011; Carter et al., 2017). Studies of the expanded 2007 RFS mandate further show that biofuel policies affected both the level (Hochman, Rajagopal, and Zilberman, 2010; Hausman, Auffhammer, and Berck, 2012; Chen and Khanna, 2013) and volatility of corn prices (Wright, 2011; Wright 2014). A parallel strand of research examined the relationship between agricultural commodity prices and energy prices, documenting significant price linkages and volatility spillovers between the two markets (Zhang et al., 2009, 2010; Serra, 2011; Serra, Zilberman, and Gil, 2011, among others). More recent studies have begun to examine the RD boom, focusing on

the soybean oil and crushing market (Avileis, 2025; Avileis, Swanson, and Irwin, 2026; Kang and Lee, 2025).

This literature, however, shares three important limitations. First, most empirical contributions focus on upstream commodity markets rather than downstream food and energy prices faced by consumers. This distinction is important because commodity price movements do not translate one-for-one into retail prices—processing margins, transportation costs, and retailer pricing behavior can absorb or amplify shocks as they move through the supply chain (Meyer and von Cramon-Taubadel, 2004; Nakamura and Zerom, 2010). Moreover, retail prices are the outcomes most directly relevant to household welfare, especially for low-income consumers. Although recent work has examined pass-through of RFS implicit subsidies to retail fuel prices (Lade and Bushnell, 2019) and impacts on beef herd size (Davis, Adjemian, and Langemeier, 2023), less is known about how it affects broader measures of domestic retail food and fuel prices.

Second, establishing the causal effects of biofuel policy is challenging because observed price movements reflect many simultaneous forces rather than isolated policy changes. Most existing studies rely on reduced-form models or partial equilibrium models. Reduced-form approaches that use biofuel production volumes or mandate levels as policy proxies face endogeneity concerns because policy decisions may respond to prevailing market and economic conditions (de Gorter and Just, 2010; Lade and Bushnell, 2019; Irwin, 2019). Partial equilibrium models, while useful for counterfactual analysis, require assumptions about market clearing, supply elasticities, and cross-market linkages that may be difficult to validate, particularly during periods of rapid market changes. These limitations motivate an identification strategy that isolates plausibly exogenous variation in biofuel policy from other contemporaneous market shocks.

Third, aside from a few recent studies (Avileis, 2025; Avileis, Swanson, and Irwin, 2026; Kang and Lee, 2025), relatively little empirical evidence exists regarding the recent RD boom. As shown in Figure 1, the rapid RD expansion is moving the center of gravity from corn ethanol, which had dominated the policy landscape since the mid-2000s but faced the blend wall constraint, toward soybean oil-based biofuel, which faces fewer blending limitations. Because

much of the existing literature focuses on the corn ethanol era, it offers limited evidence on the price effects of biofuel policy in this newer RD-centered regime.

This paper addresses these gaps by developing a high-frequency measure of biofuel policy news and using it to estimate the effects of biofuel policy shocks on energy prices, agricultural commodity prices, and downstream food price measures over 2007-2025. We construct the news series from changes in ethanol and soybean oil futures prices around regulatory announcements by the Environmental Protection Agency (EPA) and the California Air Resources Board (CARB), as well as *Reuters* reports of leaked EPA regulatory decisions. We then use the series as an external instrument in a Proxy Structural Vector Autoregression (SVAR-IV) framework (Stock and Watson, 2012; Mertens and Ravn, 2013) to estimate the dynamic effect of biofuel policy shocks.

This design draws on the logic of high-frequency identification, which uses asset price movements around well-timed announcements to isolate market-relevant policy news. The approach was pioneered in the monetary policy literature using interest rate surprises around Federal Open Market Committee (FOMC) meetings (Bagliano and Favero, 1999; Kuttner, 2001; Faust et al., 2004; Gürkaynak et al., 2005; Gertler and Karadi, 2015; Nakamura and Steinsson, 2018; Bauer and Swanson, 2023, among others) and has since been extended to oil markets (Känzig, 2021), U.S. Treasury markets (Phillot, 2025), and agricultural commodity markets (Adjemian and Jo, 2025). We bring this identification strategy to the biofuel regulatory setting, where discrete policy announcements, active futures markets, and the close integration of energy and agricultural markets create a distinct opportunity to identify policy-induced price movements.

Our baseline SVAR model includes the global real Brent oil spot price alongside a suite of U.S.-specific variables: the real biofuel price, the real field crop price, real personal consumption expenditures (RPCE) for food, the headline consumer price index (CPI), the food-at-home CPI, and the food service CPI. The latter four variables represent the key downstream price indices, capturing aggregate food expenditures, overall inflation, grocery prices, and food service costs, respectively. The real biofuel price, a production-weighted composite of ethanol and biodiesel prices, is used to measure biofuel market shocks, while the constructed biofuel news series serves as an instrument for the shock to identify the SVAR model. The sample spans November 2007 to

February 2025 at a monthly frequency. Because the sample includes the extreme macroeconomic volatility of early 2020, we follow Lenza and Primiceri (2022) and down-weight the pandemic observations when estimating the model. This prevents a handful of COVID-19 months from dominating the residual covariance structure and distorting the estimated dynamics; we describe the adjustment in Section 4.

We find that tighter biofuel policy generates inflationary pressures that pass from upstream commodity markets to downstream consumer prices. A shock normalized to raise real biofuel prices by 2 percent increases real oil and field crop prices by approximately 1.5 and 1.7 percent, respectively, creating cost pressures throughout upstream food and energy supply chains. The shock propagates to downstream markets gradually. Headline CPI increases by 0.1 percent two years after the shock, while food-at-home prices rise by 0.23 percent, consistent with direct commodity-to-retail pass-through from field-crop prices. Food-away-from-home prices increase by 0.09 percent at their peaks, reflecting differences in commodity-cost shares across these categories. Household food spending declines slightly.

The inflationary burden is not evenly distributed across food categories. Commodities most directly tied to biofuel production, particularly fats and oils and dairy, experience the sharpest price increases, while categories with limited feedstock exposure show muted responses. Producer prices respond more strongly than consumer prices across several categories, indicating that upstream cost pressures are partially absorbed before reaching consumers. Beyond food markets, the shock generates broader inflationary effects, raising retail energy prices and consumer inflation expectations. This suggests that biofuel policy affects not only realized prices but also beliefs about future inflation.

The burden also falls unevenly across households. Using a distributional consumer price index (Jaravel, 2026) constructed separately for lower-middle-, middle-, and upper-middle-income groups, we find that a biofuel policy shock raises prices more for lower-middle-income households than for upper-middle-income households. Because lower-income households devote a larger share of their budgets to food and energy, the resulting increase in their cost of living is larger, suggesting that the shock is regressive.

We also assess the quantitative importance of biofuel policy shocks using historical decompositions (HDs) and forecast error variance decompositions (FEVDs). HDs show that these shocks have contributed substantially to downstream food prices. Their contribution to the food-at-home price level reaches roughly 2 percent in absolute value in both directions over the sample. Pass-through declines sharply along the supply chain—the contribution to grocery prices is about three times larger than that of restaurant prices. For food price inflation (year over year), biofuel policy shocks added around 1.1 percentage points at the height of the 2021-22 surge. FEVDs reinforce these findings. Biofuel policy shocks account for more than 20 percent of food prices and food expenditures in the long run, a substantial share of variation in field crop and soybean oil prices, and about 14 percent of long-run variation in the headline CPI.

This paper contributes to the literature in three ways. First, we extend the focus of the food-versus-fuel literature beyond upstream agricultural commodity markets by estimating the effects of biofuel policy shocks on retail food prices and consumer food expenditures, which are central to evaluating the household-level consequences of biofuel policy. Second, we construct a novel biofuel policy news series from announcements by the EPA, CARB, and Reuters and use it as an external instrument in an SVAR-IV framework to address the endogeneity concern. Third, our sample spans both the corn ethanol mandate era and the recent RD boom, providing evidence on the price effects of biofuel policy over a period that extends well beyond the corn ethanol focus of most existing studies.

The rest of this paper proceeds as follows. Section 2 provides institutional background on U.S. biofuel policy. Section 3 describes the identification strategy and the construction of the biofuel policy news series. Section 4 presents the empirical framework. Section 5 reports the main results. Section 6 presents robustness checks, and the last section concludes.

2. Institutional Background

2.1 The Renewable Fuel Standard (RFS)

The Renewable Fuel Standard (RFS) mandates minimum annual volumes of renewable fuels in the U.S. transportation fuel supply. Established by the Energy Policy Act of 2005 and expanded under the Energy Independence and Security Act of 2007, the program was designed to reduce

lifecycle greenhouse gas emissions, strengthen domestic energy independence, and revitalize rural economies by creating stable demand for agricultural feedstocks. The statute envisioned renewable fuel use rising from 9 to 36 billion gallons in 2008-2022 (EPA, 2025*b*). In practice, the EPA-determined renewable fuel obligation for 2023 reached 20.8 billion gallons (EPA, 2023*a*).

The EPA administers the RFS through a market-based system of Renewable Identification Numbers (RINs)—tradeable compliance credits generated when biofuels are blended with petroleum-based fuels. “Obligated parties,” primarily refiners and importers, must acquire and retire sufficient RINs annually to demonstrate compliance with mandated blending volumes across four fuel categories (Gerveni and Irwin, 2025). This credit trading system provides compliance flexibility, enabling obligated parties with deficits to purchase RINs from those that generate surplus credits (Lade and Smith, 2025).

Each year, the EPA sets Renewable Volume Obligations (RVOs), which translate statutory and regulatory blending targets into percentage standards determining the number of RINs that obligated parties must acquire and retire. Proposed and final RVO announcements, therefore, convey important regulatory news because they directly affect expected demand for RINs and biofuels. Figure 2 plots RIN prices for biomass-based diesel (D4) and conventional renewable fuel (D6, primarily corn ethanol) from 2010 to 2025. The series highlights five market episodes.

Biodiesel mandate expansion (2010—2013). Following the 2007 RFS expansion, biomass-based diesel mandates grew from 0.5 to over 1.3 billion gallons between 2009 and 2013 (EPA, 2025*c*), placing upward pressure on D4 RIN prices as production capacity struggled to keep pace. Prices subsequently eased as capacity expanded and the industry became better able to meet RFS2 requirements.

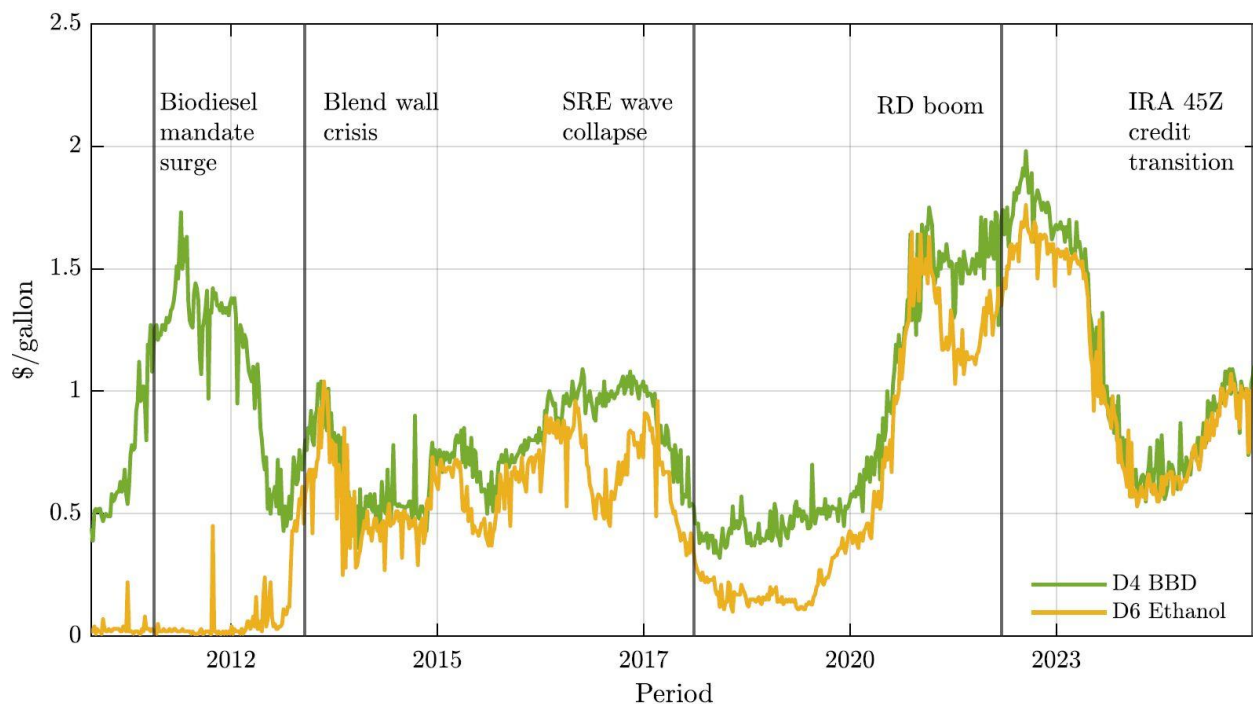
The binding blend wall (2013—2015). D6 RIN prices surged above \$1.00 per gallon around 2013 as corn ethanol approached the practical blending limit of roughly 10 percent (Lade and Smith, 2025). Unable to expand ethanol consumption further, obligated parties increasingly relied on higher-cost biodiesel to satisfy overall RFS obligations, pushing D4 RIN prices higher as well (Irwin and Good, 2013).

SRE wave collapse (2016—2019). An expanded use of small refinery exemptions effectively reduced aggregate compliance obligations, weakening RIN demand and depressing both D4 and D6 prices (Irwin, 2019). This episode illustrates the endogenous nature of biofuel policy: regulatory discretion can directly shape market conditions.

Renewable Diesel boom (2020—2023). D4 RIN prices surged to historically high levels as renewable diesel—a drop-in fuel without blend wall constraints—rapidly expanded, reshaping the biomass-based diesel market (Ramsey, 2024; Gerveni, Hubbs, and Irwin, 2024).

IRA 45Z credit transition (2025—present). The introduction of the Clean Fuel Production Credit (Section 45Z) under the Inflation Reduction Act created regulatory uncertainty, contributing to declining D4 and D6 RIN prices as producers awaited clarity on the new credit structure. As illustrated in the figure, both D4 and D6 RIN prices declined substantially during this period, consistent with market concerns about subsidy intensity and uncertainty over qualifying fuel pathways under the new framework (Buffie, 2025).

Figure 2. Biomass-Based Diesel (D4) and Renewable Fuel (D6) RIN prices, 2010-2025



Source: U.S. Environmental Protection Agency, RIN Trades and Price Information

Notes: Figure shows weekly volume-weighted average RIN prices for Biomass-Based Diesel (D4) and Renewable Fuel (D6) by transfer date, averaged across vintages when multiple production years were traded.

2.2 Low Carbon Fuel Standards (LCFS)

Complementing federal mandates, state-level Low Carbon Fuel Standards (LCFS) have emerged as significant drivers of biofuel demand, particularly for lower-carbon advanced fuels. California pioneered this approach in 2011, implementing an LCFS that initially targeted a 10 percent reduction in the carbon intensity of transportation fuels by 2020. The California Air Resources Board (CARB) has since strengthened this target to a 30 percent reduction by 2030 (CARB, 2024). Unlike the RFS's volumetric mandates, the LCFS operates on comprehensive lifecycle carbon intensity assessments, assigning credits to low-carbon fuels and deficits to high-carbon petroleum based on total supply chain emissions (CARB, 2025). Given California's substantial fuel market—representing approximately 10 percent of U.S. transportation fuel consumption—the LCFS generates spillover effects nationally, with Oregon, Washington, and New Mexico adopting similar programs in subsequent years (Bushnell et al., 2024).

CARB publishes regular LCFS compliance reports that provide information on credit generation, deficits, fuel volumes, and feedstock use. These reports can convey market-relevant information about the balance of low-carbon fuel supply and compliance demand, making them a useful source of regulatory news for identifying biofuel policy shocks.

3. Identification Strategy

This section describes our identification strategy for isolating biofuel policy news. The key challenge is that changes in biofuel price may reflect multiple simultaneous factors rather than policy changes alone. We address this by measuring biofuel futures price changes within a narrow window around regulatory announcements, when policy news is plausibly the dominant source of new information.

The institutional setting supports this approach. EPA proposed and final RVO announcements directly affect expected RFS compliance demand, while CARB's LCFS compliance reports provide information on credits, deficits, fuel volumes, and feedstock use. In addition, Reuters reports of leaked or anticipated EPA decisions may also reveal policy-relevant information

before official announcements. These discrete regulatory events, together with active futures markets for ethanol and soybean oil, allow us to construct a high-frequency measure of biofuel policy news, which we use as an external instrument in SVAR-IV as described in Section 4.

3.1 Biofuel policy announcements

We construct a database of biofuel policy announcements from three primary sources: the Environmental Protection Agency (EPA), California Air Resources Board (CARB), and *Reuters* news reports, following Avileis, Swanson, and Irwin (2026). The EPA announcements included proposed and final RVOs for the RFS.¹ Although these releases do not occur on fixed calendar dates, they are closely watched by market participants because they determine expected compliance demand for RINs and biofuels. The CARB announcements consist of quarterly LCFS compliance reports, which provide information on credit generation, deficits, fuel volumes, and feedstock consumption. These reports are market-relevant because they reveal the balance between low-carbon fuel supply and compliance demand under California's carbon-intensity standard. To capture information that may reach markets before official EPA announcements, we supplement these institutional releases with *Reuters* articles reporting leaked or anticipated EPA regulatory actions. We focus on *Reuters* as it has been a primary outlet for leaked or pre-announced information regarding EPA biofuel policy (Avileis, Swanson, and Irwin, 2026).

Because CARB LCFS compliance reports are released at 3 pm PST, after market close, the relevant trading day is the following business day. Similarly, *Reuters* articles published outside of trading hours are assigned to the following business day. When multiple biofuel announcements occur within a two-day window, we retain only the earliest announcement to avoid double-counting. Our final sample comprises 98 biofuel policy announcements from November 2007 to February 2025. A complete list of announcements is provided in Appendix A.

¹ EPA biofuel news are obtained from the 'News, Notices and Announcements for the Renewable Fuel Standard' EPA website: <https://www.epa.gov/renewable-fuel-standard/news-notice-and-announcements-renewable-fuel-standard>. There are different types of announcements. We focus on the publication date of 'Proposed Renewable Fuel Standards' and 'Final Renewable Fuel Standards' related to the supply of biofuel energy (see Avileis, Swanson, and Irwin, 2026, for details).

3.2 Biofuel futures markets

Ethanol and biomass-based diesel are the two primary biofuel commodities relevant to RFS compliance. For ethanol, we consider the Chicago Ethanol (Platts) Futures contract, a financially settled derivative traded on the New York Mercantile Exchange (NYMEX). It is widely regarded as the most liquid ethanol derivative in the United States (Gerveni et al., 2023), making it a central instrument for price discovery and risk management in the North American biofuel market.

For biomass-based diesel, a direct futures contract would be the natural proxy for price dynamics. However, the CME biodiesel contract, which began trading in 2011, lacks both sufficient price history and trading liquidity for our full sample. We therefore use soybean oil futures as a proxy for biomass-based diesel market conditions. Soybean oil is the dominant feedstock for U.S. biodiesel and RD production, accounting for about 55 percent of total feedstock inputs (Gerveni, Hubbs, and Irwin, 2024). Because soybean oil and biodiesel prices are closely linked through this feedstock relationship (Irwin and Good, 2017), soybean oil futures provide a reasonable proxy over the sample period. The CBOT Soybean Oil futures contract is also among the most actively traded agricultural commodity derivatives globally.

3.3 Identification of biofuel policy news

To isolate exogenous variation in biofuel energy prices, we employ a high-frequency identification strategy following Känzig (2021). We measure biofuel policy news as the log change in biofuel futures prices within a daily window around each announcement:

$$IndividualBiofuelPolicyNews_{m,d,i}^h = 100 \times (\ln F_{m,d,i}^h - \ln F_{m,d-1,i}^h), \quad (1)$$

where d , m , and i are the day, announcement month, and biofuel prices (ethanol or soybean oil), respectively. $F_{m,d}^h$ represents the settlement price for h -deferred futures contract of a commodity. Here, we consider $h=1,2$, representing the nearby and the first-deferred futures contracts, respectively.²

For each commodity, we aggregate individual biofuel policy news across the nearby and first-deferred futures contracts using the first principal component (Pearson, 1901; Jolliffe and

² A constant risk premium is assumed within the window.

Cadima, 2016), which accounts for 78 percent of the variation across the two contract maturities. We then construct a single biofuel policy news measure by weighting each commodity's series based on monthly production shares of ethanol and soybean oil-based biomass-diesel:

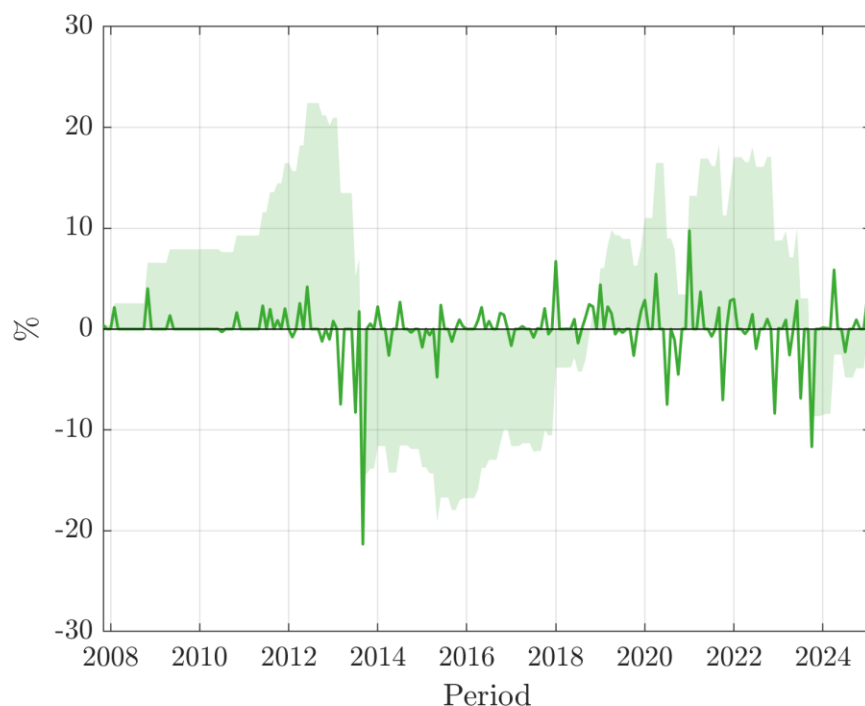
$$BiofuelPolicyNews_{m,d}^h = \sum_i IndividualBiofuelPolicyNews_{m,d,i}^h \times S_{m,i} \quad (2)$$

where $S_{m,i}$ indicates the production share of commodity i in month m . The weighted series represents our biofuel policy news measure.

Because the economic variables in our baseline model are observed at a monthly frequency, we aggregate the daily biofuel policy news series, $BiofuelPolicyNews_{m,d}^h$, into a monthly series. The monthly value equals the announcement-day news in months with a single announcement, the sum of announcement-day news in months with multiple announcements, and zero in months without any announcements. This convention follows Känzig (2021) and reflects the assumption that all announcements within a given month contribute independently to the total policy news during that period. Summing rather than averaging preserves the total information content of the monthly policy signal, while assigning zero to months without announcements correctly captures the absence of new regulatory information.

Figure 3 illustrates the historical series of the U.S. biofuel policy news from 2007 to 2025. The figure shows that biofuel policy news routinely surprises the markets, especially in the mid-2010s (2013-2015) and after 2020. These periods coincide with major expansions in biofuel production capacity. In contrast, during the mid-2010s (2016-2019), market surprises were more muted as the U.S. government granted numerous small refinery exemptions that reduced net compliance obligations, weakening the policy's aggregate demand impact on biofuels (Coppess and Irwin, 2018; Irwin, 2019).

Figure 3. Biofuel policy news series, November 2007- February 2025



Source: Bloomberg terminal and authors' calculations

Notes: The biofuel policy news series is constructed as the first principal component of changes in the U.S. biofuel energy futures (ethanol and soybean oil; relative to their share of domestic production). The solid line represents the monthly biofuel policy news series, and the shaded area illustrates the cumulative series, which is the sum of the monthly news series over time.

3.4 Diagnostic tests of biofuel policy news

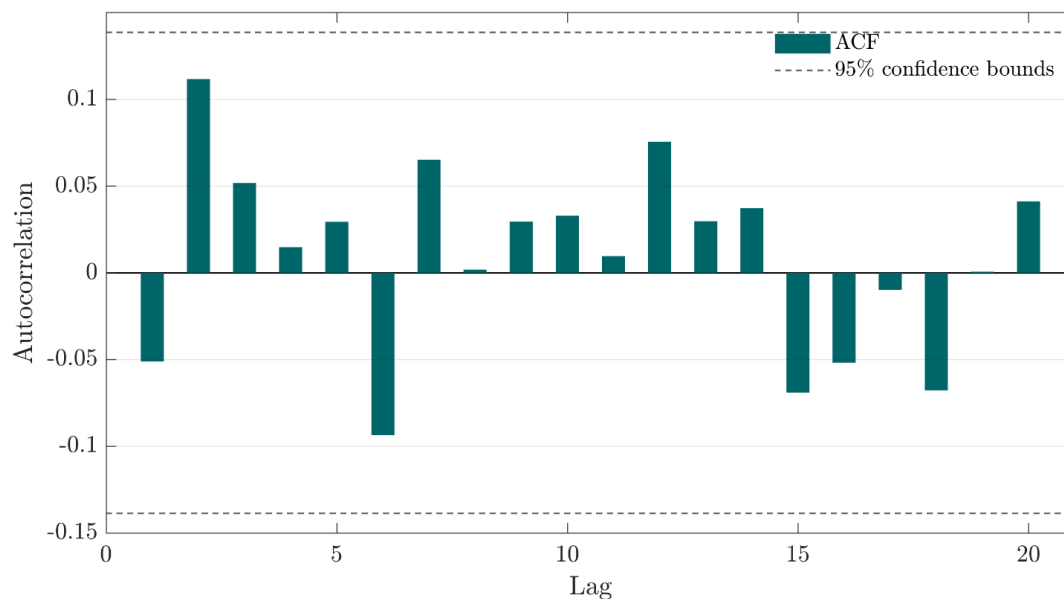
Before employing the biofuel policy news series as an instrument, we verify its validity through three diagnostic tests: autocorrelation, predictability, and orthogonality to external shocks. Figure 4 shows that the instrument is not predictable from its own past. Table 1 reports Granger causality test results, showing that neither lagged baseline model variables nor biofuel production predict the news series. Table 2 further shows that the instrument is largely uncorrelated with a broad set of external shocks and indices.

We additionally evaluate informational sufficiency following Mori and Peersman (2026). A valid proxy should not be predictable from information available to market participants beyond that already embedded in the VAR. We therefore examine whether five information blocks—financial and credit conditions, policy uncertainty, commodity and global factors, monetary policy, and weather—predict the news series, the reduced-form residual of the real biofuel price

equation, and the identified structural shock. Each object is first orthogonalized with respect to twelve lags of the baseline VAR variables, allowing the test to isolate predictive information not already embedded in the model. Appendix C reports the testing results.

Over the full sample, none of the five blocks predicts the structural shock, and the reduced-form residual yields the same conclusion. These results support informational sufficiency with respect to the standard set of potential confounders. In the pre-2020 subsample, however, the policy-uncertainty block is marginally significant, suggesting that it may contain information about earlier biofuel regulatory decisions. We thus conduct a robustness check using an instrument orthogonalized with respect to the policy-uncertainty block, as discussed in section 6.1.

Figure 4. Autocorrelation test



Source: Bloomberg terminal and authors' calculations

Notes: The pink horizontal dashed lines represent the 95 percent confidence intervals.

Table 1. Granger causality test

Variables	P-value
Biofuel policy news	0.9469
Real biofuel price	0.3730
Real oil price	0.9530
Real field crop price	0.3772

RPCE for food	0.3658
CPI	0.2044
Food-at-home price	0.9289
Food service price	0.9548
Ethanol production	0.2890
Biodiesel production	0.7302
All	0.4731

Notes: The table reports p-values from Granger causality tests of whether each variable Granger-causes the biofuel policy news series. The lag order is 12, consistent with the baseline VAR specification. “All” reports the p-value for the joint test that all variables have not predictive power for the news series.

Table 2. Correlation test with external shocks

External Shocks	Correlation	P-value
Agricultural supply news shocks	-0.00	0.96
Oil supply news shocks	0.04	0.55
Oil supply shocks	0.05	0.45
Oil consumption demand shocks	0.07	0.31
Oil inventory demand shocks	-0.04	0.60
EU carbon policy shocks	-0.03	0.71
Climate policy uncertainty shocks	-0.07	0.42
BRW's monetary policy shocks	0.06	0.39
BS's monetary policy shocks	0.05	0.51
Global supply chain pressure	0.04	0.55
Geopolitical risk	-0.04	0.57
Trade policy uncertainty	0.07	0.33
Economic policy uncertainty	0.04	0.60

Notes: The table reports correlations between biofuel policy news series and external shocks: agricultural supply news shocks (Adjemian and Jo, 2025); oil supply news shocks (Känzig, 2021); oil supply, oil consumption demand, and oil inventory demand (Baumeister and Hamilton, 2019); EU carbon policy shocks (Känzig, 2023); climate policy uncertainty shocks (Gavrilidis et al., 2026); monetary policy shocks from Bu, Rogers, and Wu (BRW; 2021) and Bauer and Swanson (BS; 2023); the Global Supply Chain Pressure Index (Federal Reserve Bank of New York); the geopolitical risk index (Caldara and Iacoviello, 2021); and trade policy uncertainty and economic policy uncertainty indices (Baker et al., 2016).

4. Empirical Approach and Data

Having constructed and validated the biofuel policy news series, we use it as an instrument in an SVAR-IV framework, following Stock and Watson (2012) and Mertens and Ravn (2013), to estimate the impact of biofuel policy shocks. This approach addresses the endogeneity of observed biofuel price movements, which may reflect policy actions as well as broader macroeconomic, agricultural, and energy-market conditions. The instrument isolates the policy-driven component of these movements by requiring the news series to be correlated with the structural shock of interest (“relevance”) but orthogonal to other disturbances (“exogeneity”). Unlike a direct shock measure, the instrument need not perfectly capture the structural shock itself. This is particularly important because the series captures only part of regulatory activity, is subject to measurement error (see Stock and Watson, 2018), and may have limited statistical power (see Nakamura and Steinsson, 2018).

4.1 SVAR model

We specify a reduced-form VAR (p) model:

$$y_t = \alpha + A_1 y_{t-1} + \dots + A_p y_{t-p} + u_t, \quad (3)$$

where p is the lag order, y_t is an $n \times 1$ vector of endogenous variables comprising the real biofuel price, real oil price, real field crop price, RPCE for food, CPI, food-at-home CPI, and food-away-from-home CPI, u_t is an $n \times 1$ vector of reduced-form innovations with covariance matrix $Var(u_t) = \Sigma$, α is an $n \times 1$ vector of constants, and $A_1, \dots, A_p$ are $n \times n$ coefficient matrices.

The reduced-form innovations in (3) are assumed to be linear combinations of structural shocks:

$$u_t = B \varepsilon_t, \quad (4)$$

where B is a non-singular $n \times n$ structural impact matrix and ε_t is a $n \times 1$ vector of mutually uncorrelated structural shocks (i.e., $Var(\varepsilon_t) = \Omega$ is diagonal). From this mapping,

$$\Sigma = B \Omega B'. \quad (5)$$

We order the biofuel policy news shock as the first structural shock (i.e., $\varepsilon_{1,t}$), reflecting the assumption that biofuel policy announcements respond to regulatory and political processes

rather than to contemporaneous movements in energy, commodity, or consumer prices.³ Our focus is thus on identifying the structural impact vector b_1 or the first column of B , which allows us to determine how biofuel market news shocks affect the other variables in the system.

4.2 External instrument approach

Full identification of B requires n^2 restrictions, while Equation (5) provides only $n(n + 1)/2$ unique restrictions. Rather than imposing additional zero or sign restrictions, we follow Stock and Watson (2012) and Mertens and Ravn (2013) and use an external instrument to achieve partial identification of b_1 alone.

Let z_t denote the instrument, namely the biofuel policy news. Identification of b_1 , requires instrument relevance and exogeneity:

$$\mathbb{E}[z_t \varepsilon_{1,t}] = \theta \neq 0, \quad (6)$$

$$\mathbb{E}[z_t \varepsilon_{2:n,t}] = 0, \quad (7)$$

where $\varepsilon_{1,t}$ is the biofuel policy news shock and $\varepsilon_{2:n,t}$ is the $(n - 1) \times 1$ vector of other structural shocks. Equation (6) requires the news series to be correlated with shifts in biofuel policy stringency, while Equation (7) requires it to be uncorrelated with contemporaneous shocks of other variables. Under these two conditions, b_1 is identified up to a scaling factor (Stock and Watson, 2018):

$$\tilde{b}_1 \equiv \theta_0 \propto \mathbb{E}[z_t u_{2:n,t}] / \mathbb{E}[z_t u_{1,t}], \quad (8)$$

where θ_0 indicates unit-effect normalization and $\mathbb{E}[z_t u_{1,t}] \neq 0$.⁴

³ Note that this ordering refers to the structural shocks, not the endogenous variables in the VAR system. The first endogenous variable in y_t is the real biofuel price—a production-weighted composite of ethanol nearby futures and biodiesel spot prices—which serves as the observable measure of biofuel market conditions. The structural shock is normalized so that it raises the real biofuel price by 2 percent.

⁴ Unit-effect normalization fixes the scale of $\varepsilon_{1,t}$ such that a unit increase in the structural shock leads to a one-unit increase in $y_{1,t}$. For instance, a 1 percent biofuel policy shock raises the biofuel price by 1 percent. This normalization has an important advantage over the more common unit standard deviation normalization: it allows direct estimation of dynamic causal effects in policy relevant units, making the impulse responses immediately interpretable without additional scaling (see Stock and Watson, 2018, for details).

With unit-effect normalization, we scale the structural shock $\varepsilon_{1,t}$ such that it corresponds to a 2 percent increase in the real biofuel price, allowing the impulse responses to be interpreted directly in policy-relevant units (Stock and Watson, 2018):

$$\tilde{b}_1 \equiv \mathbb{E}[z_t u_{2:n,t}] / \mathbb{E}[z_t u_{1,t}] \times \omega. \quad (9)$$

4.3 COVID-era volatility weighting

Our sample spans the COVID-19 period, during which unusually large macroeconomic disturbances could dominate the reduced-form covariance matrix and distort the estimated impulse responses. Following Lenza and Primiceri (2022), we model the reduced-form residuals as $u_t = s_t \cdot \varepsilon_t$, where ε_t is homoskedastic with covariance Σ and the scale s_t equals one outside the pandemic window. We allow s_t to take separate values in March 2020, April-May 2020, and June-August 2020, and to decay geometrically at rate ρ thereafter. These parameters are estimated by maximum likelihood. The estimates place a scale of 18.8 on March 2020 and unity on all subsequent months ($s = 1.0$ for April-May and June-August 2020). Because the post-summer scale equals one, ρ is not separately identified. In practice, the adjustment therefore amounts to heavily down-weighting March 2020—equivalent to including an indicator for that month—so that the VAR, estimated by weighted least squares with weights $1/s_t$, is not distorted by the volatility during the pandemic.

4.4 Variables and Data

The baseline model includes seven variables to capture the full transmission pathway from biofuel policy shocks through energy and commodity markets to downstream consumer prices. Real biofuel price, a production-weighted composite of the ethanol nearby futures price and the biodiesel spot price, serves as the primary policy-sensitive variable through which biofuel mandates transmit to energy markets and is used to normalize the structural shock. Real oil prices are included because biofuels and petroleum-based fuels are complements and substitutes in the transportation fuel market, making oil prices a key channel through which biofuel policy shocks propagate to broader energy costs. Real field crop price, a production-weighted average of corn and soybean prices, captures upstream feedstock market responses, reflecting competition for

agricultural commodities between food and fuel uses. RPCE for food and three CPI indices—headline CPI, food-at-home CPI, and food-away-from-home CPI—measure downstream pass-through to consumer prices and expenditures, allowing us to trace the complete transmission from regulatory mandates to household purchasing power.

The main sample spans from November 2007 to February 2025. The starting date is the first EPA announcement regarding the RFS, while the ending date reflects the availability of biodiesel spot price data from USDA Bioenergy Statistics, which is used to construct the real biofuel price variable in the baseline model. Appendix Table B provides full variable definitions, including those used in the extended and distributional analyses, and Table 3 reports summary statistics for the baseline variables.

Table 3. Summary statistics, November 2007-February 2025

Variable	Obs.	Mean	Std. Dev.	Min	Max
Biofuel price (cents/gallon)	208	202.88	49.27	115.44	324.60
Oil price (cents/barrel)	208	78.63	24.51	26.85	113.59
Field crop price (cents/bushel)	208	637.36	154.39	445.73	1011.94
Consumer Price Index (CPI)	208	249.82	30.33	210.83	319.68
RPCE for food (Bil. Chained 2017 \$)	208	996.77	127.18	835.70	1285.40
Food-at-home price index	208	246.39	28.39	205.12	310.96
Food-away-from-home price index	208	272.73	45.86	209.85	377.00

Notes: The biofuel price is a production-weighted composite of the nearby ethanol futures price and biodiesel spot price. The field crop price is a production-weighted composite of nearby corn and soybean futures prices, in cents per bushel. For estimation, all variables are transformed to natural logarithms, and biofuel price, oil price, and field crop price are deflated by the CPI. Details on variable descriptions and data sources are provided in Appendix B.

Sources: Bureau of Economic Analysis (BEA), Bureau of Labor Statistics (BLS), Federal Reserve Economic Data (FRED), Bloomberg terminal, USDA-ERS, USDA-NASS, and USDA Bioenergy Statistics

5. Results

This section discusses empirical findings, including IRFs, historical decompositions, FEVDs, heterogeneity across food categories, and distributional effects. Before turning to these findings,

we assess instrument strength using the first-stage F-statistic, since weak correlation between the instrument and the structural shock can bias IV estimates (Stock, Wright, and Yogo, 2002). The first-stage robust F-statistic for our composite model is 13.92, well above the conventional threshold, indicating that weak instrument concerns are limited. Instrument exogeneity, or the news series is uncorrelated with other structural shocks, is supported by the diagnostic tests reported in Section 3.4.

5.1 Impulse response functions

Figure 5 displays impulse response estimates of the baseline variables to a biofuel policy shock, normalized to generate a two percent increase in the real biofuel price, corresponding to a tightening of policy stringency. The vertical axis represents percentage changes, as all variables are expressed in natural logarithms. In addition to point estimates (solid lines), we also plot the 68% (dark) and 90% (light) confidence intervals constructed from a residual-based moving-block bootstrap with 10,000 bootstrap replications, following Jentsch and Lunsford (2022). We use a lag order of 12, consistent with the monthly data frequency.

Figure 5 shows that real oil prices rise by around 1.5 percent at their peak, suggesting that higher biofuel compliance costs are transmitted to petroleum markets through the blending relationship. In response to the shock, real field crop prices increase by about 1.7 percent, more than twice the direct feedstock transmission of 0.6-0.8 percent calculated by a simple back-of-the-envelope calculation.⁵ This amplification likely reflects acreage competition between food and fuel crops and the low short-run supply elasticity in agricultural markets (Roberts and Schlenker, 2013).

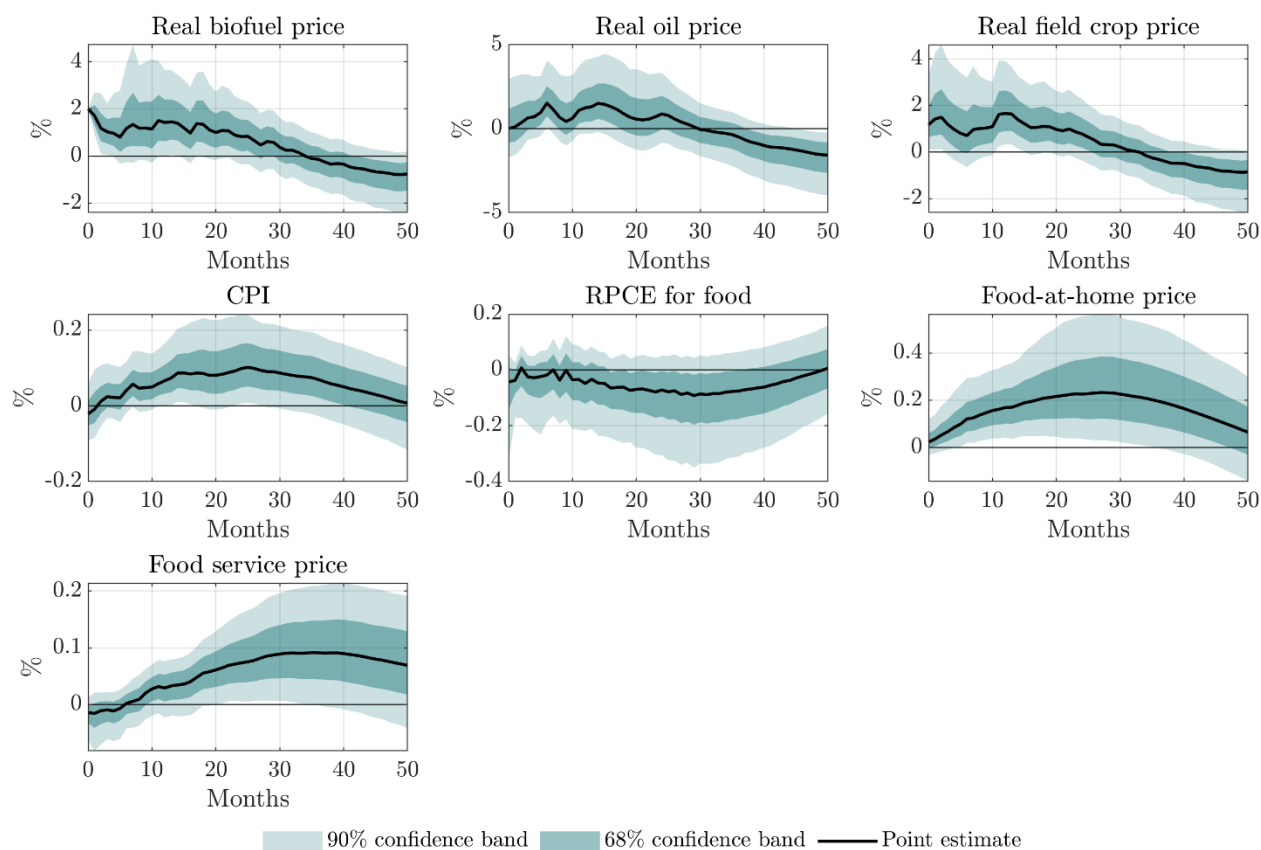
Beyond the agricultural and energy sectors, the biofuel policy shock transmits to broader cost-of-living increases that bear directly on consumers. The shock raises the headline CPI by 0.09 percent after one year. Food-at-Home CPI increases by 0.23 percent at its peak, closely aligning with the theoretical pass-through from field crop prices to grocery prices—a 1.7 percent field

⁵ The direct effect is calculated as the biofuel price shock (2%) multiplied by the average feedstock usage share for corn and soybean oil (30%-40%), yielding an expected direct price transmission of approximately 0.6%-0.8%. The observed impulse responses of field crops, 1.7%, represents an amplification factor of around 2.43 times relative to this baseline calculation.

crop price increase multiplied by the 13.4 percent average share of agricultural commodities in final food and beverage products yields an expected grocery price increase of approximately 0.23 percent (ERS, 2026). Food-Away-From-Home CPI rises by a more modest 0.09 percent, reflecting the lower commodity-cost share in restaurant meals. Real personal consumption expenditures on food decline slightly, reaching a trough of about 0.09 percent.

This contraction in real food expenditure may reflect the broader inflationary effects of the shock. By raising energy, food, and overall prices simultaneously, tighter biofuel policy reduces household purchasing power and forces consumers to reallocate spending across several increasingly expensive categories. These dynamics suggest that the welfare effects of biofuel policy extend beyond direct commodity pass-through, particularly for households that devote larger shares of their budgets to food and energy.

Figure 5. Impulse response functions to a biofuel policy shock



Notes: The shock is normalized to a two percent increase in the real biofuel price (first variable, upper left). Impulse responses are plotted over a fifty-month horizon. Confidence intervals are derived from the moving-block bootstrap with 10,000 replications.

5.2 The contribution of biofuel policy shocks to downstream food markets

Having confirmed the transmission of biofuel policy shocks through agri-food markets using IRFs, we next use historical decompositions to quantify the extent to which these shocks contributed to observed downstream price (and expenditure) movements. The contribution is obtained by propagating the identified biofuel policy shocks through the estimated VAR dynamics and cumulating their effects over time. Because we only identify one structural shock, the decomposition is partial: the remaining variation reflects all other shocks and initial conditions.

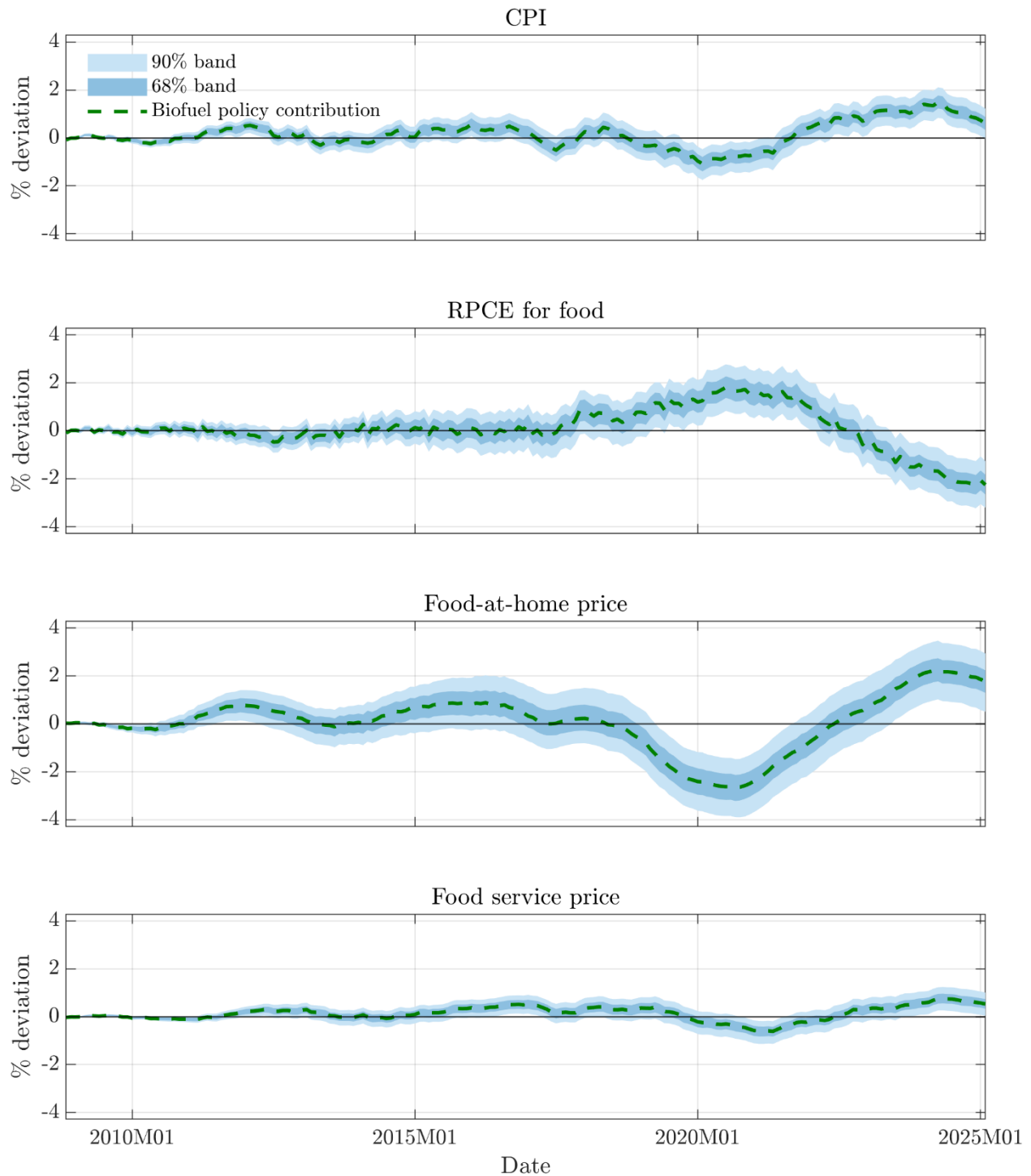
Figure 6 displays the contribution of biofuel policy shocks to four downstream price series in the baseline model: the headline CPI, real personal consumption expenditures on food, the food-at-home price index, and the food service price index. All series are in log levels, so contributions are expressed as percent deviations from the path that would have prevailed absent biofuel policy shocks.

The estimated contributions are economically meaningful and persistent. From 2019 through 2021, a period that included the expansion of small refinery exemptions and a weakening of effective RFS stringency, biofuel policy shocks kept food-at-home prices below their counterfactual path, with the gap reaching 2.7 percent in September 2020. The sign reverses from July 2023 through the end of the sample: following EPA's June 2023 Set Rule, which raised biomass-based diesel volumes relative to the proposed rule (EPA, 2023*b*), the contribution peaks at 2.3 percent in April 2024. Both episodes are statistically distinguishable from zero at the 90 percent level.

Pass-through declines sharply along the food supply chain. The peak contribution to food-at-home prices (2.3 percent) is three times the corresponding contribution to food service prices (0.8 percent). At the trough, the difference is even larger: 2.7 percent for food-at-home prices compared with 0.6 percent for food service prices. The contribution is also far more variable for food-at-home prices, with a standard deviation of 1.2 percent, compared with 0.3 percent for food service prices. This ordering aligns with the cost structure of the two series: food-at-home prices are more closely linked to agricultural commodity costs, while the other is dominated by labor, rent, and other non-commodity inputs. The contribution to the headline CPI is intermediate

in magnitude (peaking at 1.5 percent), consistent with the broader composition of the aggregate consumption basket and the presence of both food and nonfood transmission channels.

Figure 6. Contribution of biofuel policy shocks to downstream prices

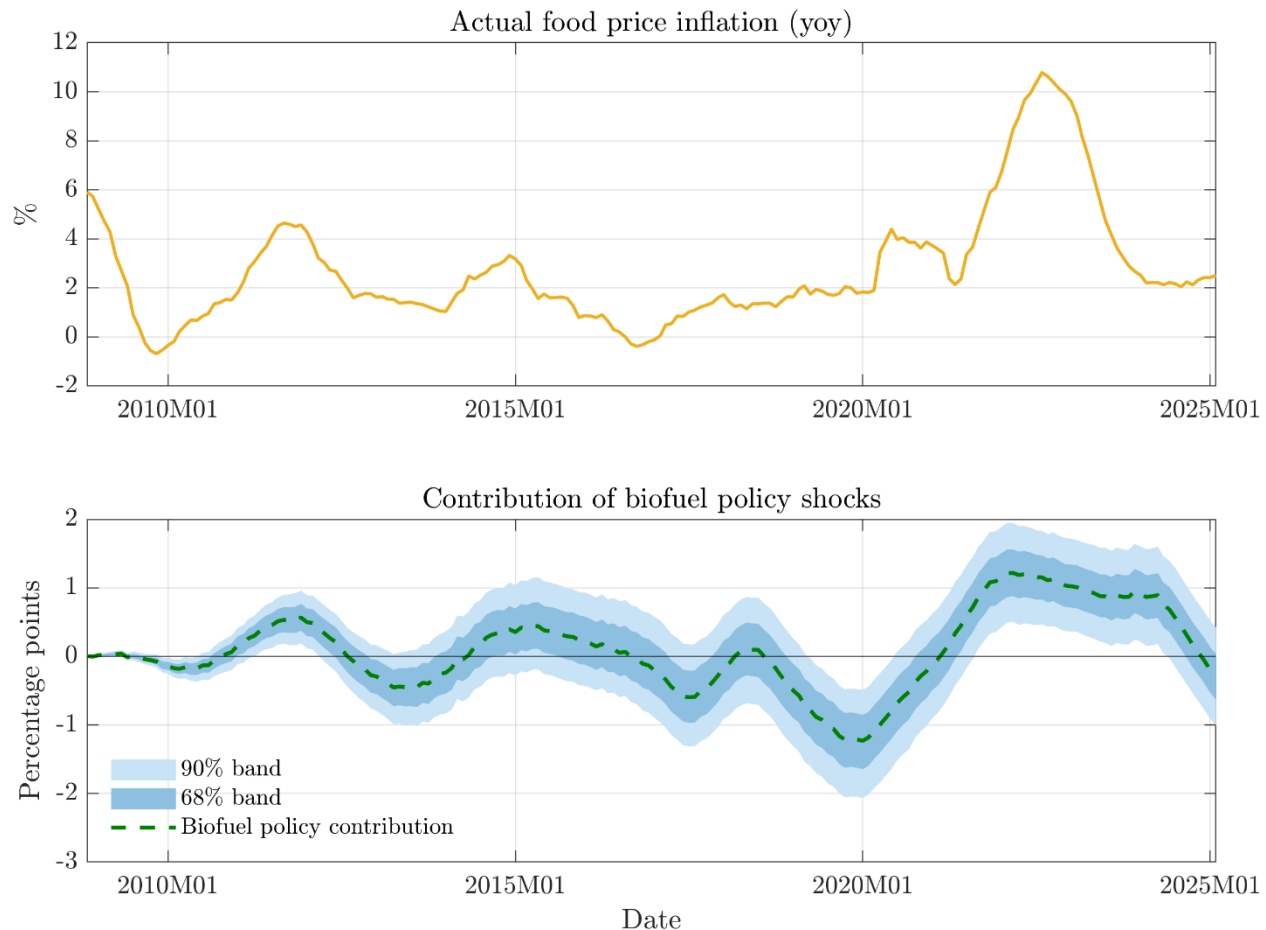


Notes: The figure shows the cumulative historical contribution of biofuel policy shocks to four downstream price series from November 2008 to February 2025, expressed as percent deviations from the counterfactual path that would have prevailed absent these shocks.

The level contributions in Figure 6 summarize the accumulated effects of past shocks and are therefore informative about the price level. Because policy debate centers on food price inflation (i.e., the rate of price change), we repeat the exercise for the year-over-year food price inflation rate, augmenting the baseline model with this series. Subtracting the estimated contribution from the realized series yields the counterfactual inflation rate that would have prevailed had biofuel policy shocks been absent, holding all other components unchanged.

Figure 7 plots the realized food price inflation in the upper panel and the estimated contribution of biofuel policy shocks in the lower panel. The shocks made a material contribution to the 2021-22 surge in food inflation, adding between 1.03 and 1.22 percentage points and remaining statistically significant throughout the period. At the August 2022 peak, food price inflation reached 10.78 percent; absent biofuel policy shocks, it would have been 9.63 percent. Biofuel policy shocks thus accounted for roughly one-tenth of peak inflation—an economically meaningful contribution, though not the dominant driver of an episode shaped primarily by historically large fiscal stimulus, supply-chain disruptions, and energy price shocks (Adjemian, Li, and Jo, 2026).

Figure 7. Contribution of biofuel policy shocks to food price inflation



Notes: The upper panel shows realized year-over-year food price inflation. The lower panel shows the contribution of biofuel policy shocks, in percentage points.

5.3 Forecast error variance decomposition

To quantify the relative importance of biofuel policy shocks in explaining variation across variables, we conduct a FEVD. To provide a broader assessment of how these shocks propagate through related markets, we extend the analysis to include two supplementary variables not contained in the baseline model: soybean oil prices and the U.S. exchange rate. Soybean oil is a major biofuel feedstock, while the exchange rate may amplify or dampen the domestic price effects of commodity shocks through import-price pass-through. Table 4 reports the share of forecast error variance attributable to biofuel policy shocks for both the baseline and supplementary variables.

Biofuel policy shocks explain 46 percent of real biofuel price variation, with the share declining to around 21 percent in the long run. The shocks also explain a substantial portion of variation in RPCE for food, food-at-home CPI, and food-away-from-home CPI, reaching 27 percent after three years, 31 percent after two years, and 30 percent after four years, respectively. This pattern is consistent with gradual pass-through through supply chains and retail pricing.

The shocks explain a modest share of field crop and soybean oil price variation, starting from 16 percent on impact and rising to 18 percent and 25 percent, respectively, after two years. The contribution to variation in headline CPI is negligible in the short run but grows over time. Variation in real oil prices and the exchange rate is barely explained by biofuel policy shocks throughout the sample horizon.

Table 4. Forecast error variance decomposition

Month	Biofuel price	Oil price	Field crop price	Soybean oil price	CPI
0	0.46 [0.01, 0.80]	0.01 [0.00, 0.13]	0.16 [0.00, 0.53]	0.16 [0.00, 0.51]	0.01 [0.00, 0.08]
12	0.26 [0.08, 0.49]	0.05 [0.01, 0.12]	0.14 [0.03, 0.39]	0.26 [0.05, 0.63]	0.03 [0.01, 0.09]
24	0.25 [0.09, 0.50]	0.07 [0.02, 0.22]	0.18 [0.05, 0.47]	0.25 [0.06, 0.62]	0.05 [0.02, 0.12]
36	0.22 [0.08, 0.45]	0.07 [0.02, 0.20]	0.16 [0.05, 0.41]	0.21 [0.05, 0.53]	0.12 [0.03, 0.37]
48	0.21 [0.08, 0.45]	0.09 [0.03, 0.24]	0.16 [0.05, 0.41]	0.20 [0.05, 0.51]	0.14 [0.03, 0.44]
Month	RPCE for food	Food-At-Home price	Food-Away-From-Home price	Exchange rate	
0	0.13 [0.00, 0.50]	0.02 [0.00, 0.16]	0.10 [0.00, 0.44]	0.00 [0.00, 0.00]	
12	0.08 [0.02, 0.24]	0.23 [0.03, 0.66]	0.08 [0.02, 0.32]	0.05 [0.01, 0.20]	
24	0.23 [0.05, 0.58]	0.31 [0.05, 0.69]	0.23 [0.04, 0.60]	0.06 [0.02, 0.20]	
36	0.27 [0.06, 0.65]	0.29 [0.04, 0.70]	0.29 [0.04, 0.71]	0.06 [0.02, 0.18]	

48	0.25	0.28	0.30	0.07
	[0.05, 0.62]	[0.04, 0.67]	[0.04, 0.72]	[0.02, 0.21]

Notes: The table reports the share of forecast error variance attributable to biofuel policy shocks at horizons of 0, 12, 24, 36, and 48 months. Numbers in brackets denote 90 percent confidence intervals constructed from 10,000 bootstrap replications.

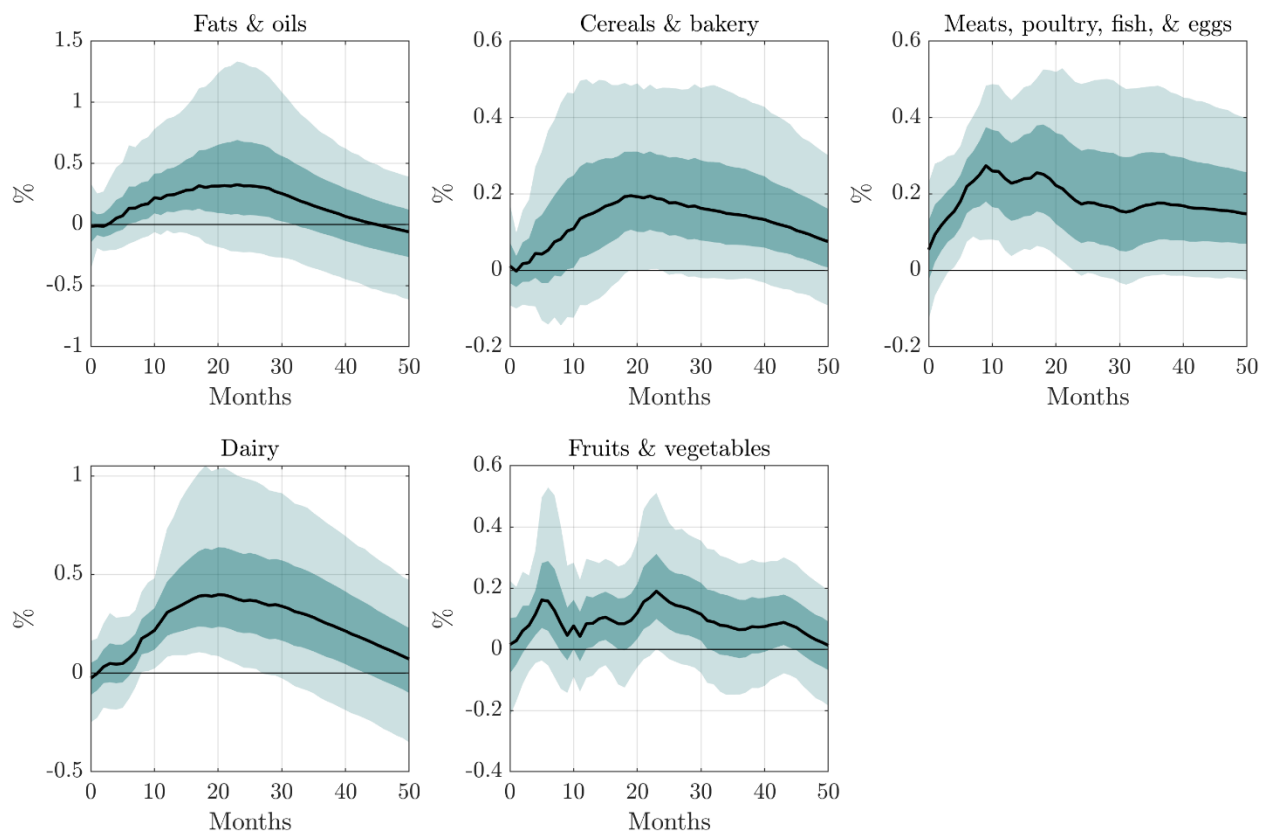
5.4 Heterogeneous price responses and broader inflationary effects

The baseline results confirm that biofuel policy shocks significantly affect agri-food markets, energy prices, consumer expenditures, and general consumer price indices. To further characterize the transmission mechanism, we analyze effects on granular food price components, retail energy prices, and inflation expectations. Following Känzig (2021), we estimate additional impulse responses by augmenting the baseline SVAR with one variable at a time. Figures 7 and 8 present impulse responses of food price subcategories for the CPI and Producer Price Index (PPI), respectively, allowing us to compare consumer- and producer-level price effects and assess how inflationary pressures are transmitted along the supply chain.

Figure 8 shows that food prices generally rise following the shock, but the magnitudes vary substantially across categories. Fats and oils prices increase by around 0.33 percent on impact, reflecting feedstock competition between food and fuel uses of soybean oil and corn (Lade and Smith, 2025). Dairy prices exhibit the largest response, rising by around 0.4 percent, followed by meats, poultry, fish and eggs at about 0.27 percent. This suggests that stronger biofuel demand raises not only direct feedstock costs but also livestock production costs by diverting corn from animal feed markets (Schmit, Verteramo, and Tomek, 2009). Cereals and bakery and fruits and vegetables prices also increase, peaking at 0.2 and 0.19 percent, respectively.

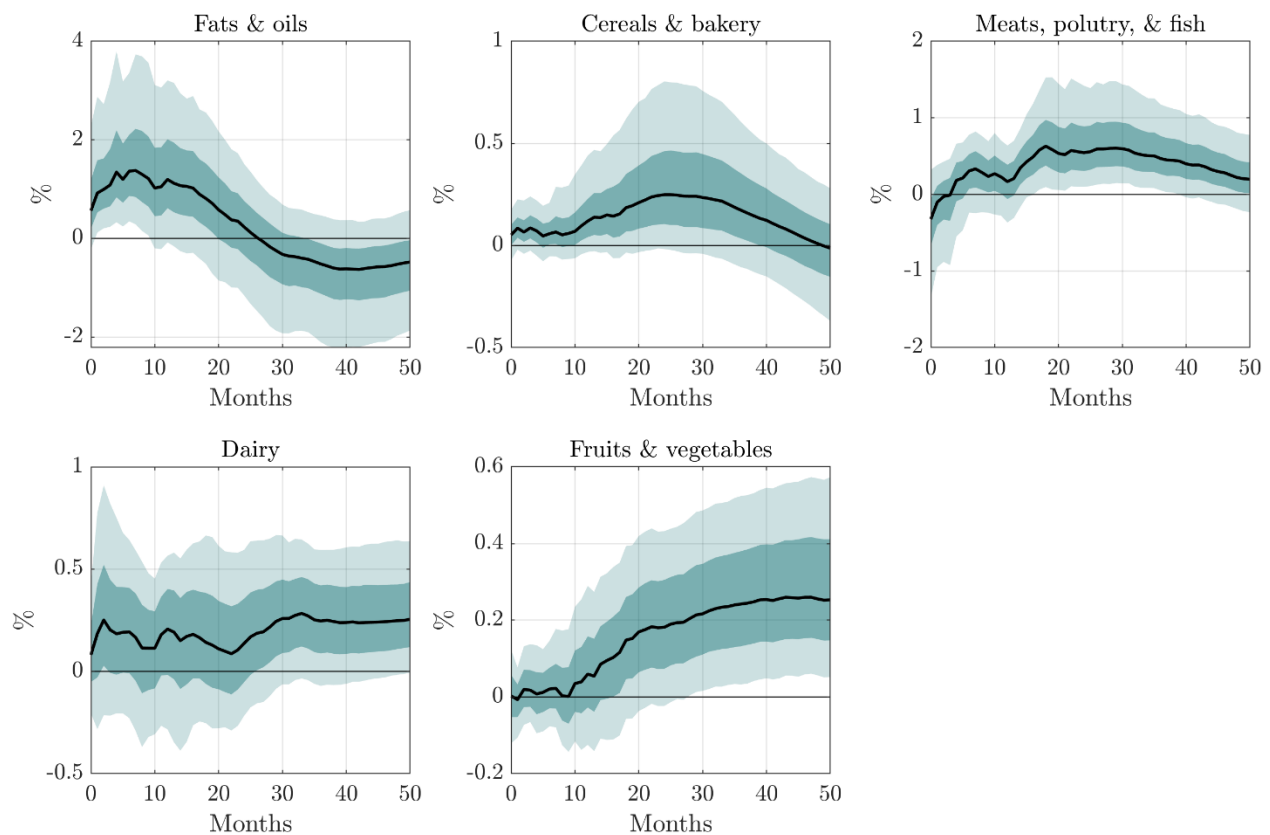
As shown in figure 9, producer prices respond more strongly than their CPI counterparts, particularly for fats and oils. The PPI for fats and oils rises by about 1.4 percent on impact, more than four times the corresponding CPI response, suggesting that processors and retailers absorb part of the upstream cost pressure, so transmission weakens as the shock moves down the supply chain toward consumers.

Figure 8. Impulse responses of food price subcategories (CPI)



Notes: The shock is normalized to a 2 percent increase in the real biofuel price. Bands as in Figure 5. Impulse responses are estimated by augmenting the baseline VAR with each Consumer Price Index (CPI) food price subcategory one at a time. Data are from the Bureau of Labor Statistics.

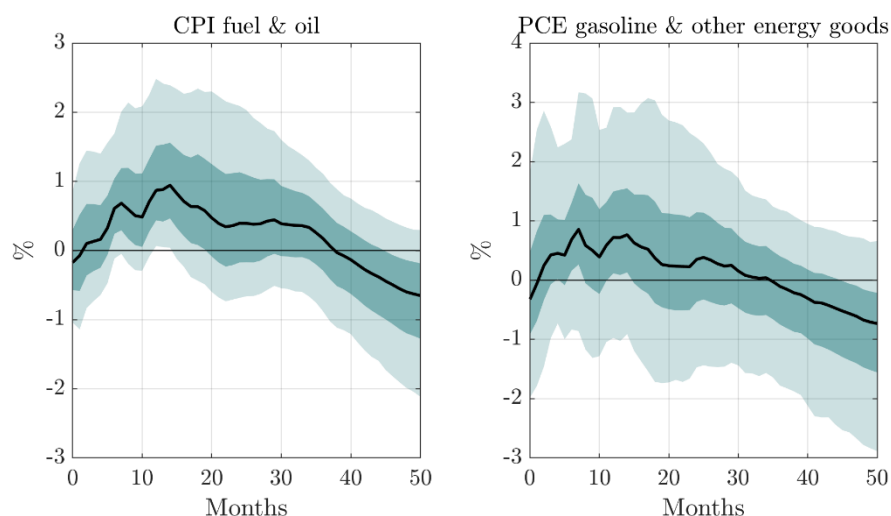
Figure 9. Impulse responses of producer price index (PPI) food subcategories



Notes: The shock is normalized to a 2 percent increase in the real biofuel price. Bands as in Figure 5. Impulse responses are estimated by augmenting the baseline VAR with each Producer Price Index (PPI) food price subcategory one at a time. Data are from U.S. Bureau of Labor Statistics.

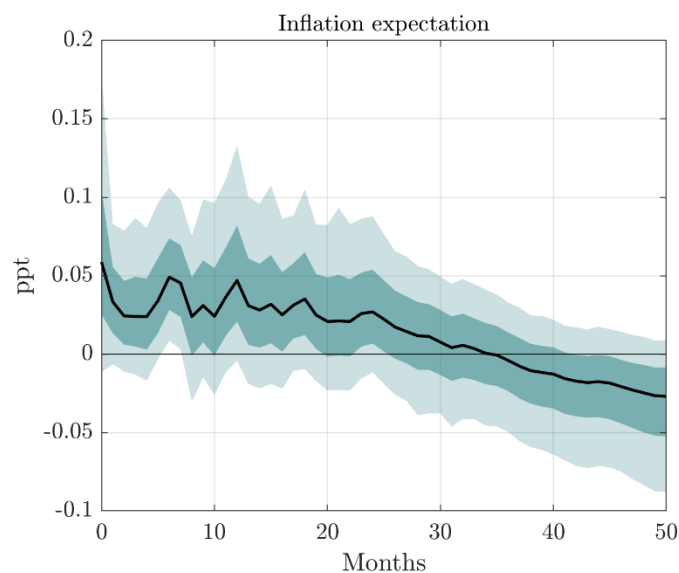
Figure 10 displays impulse responses of retail energy prices. CPI fuel and oil prices (left panel) and PCE gasoline prices (right panel) rise with a lag, although the responses are only statistically significant at the 68 percent confidence level. This pattern indicates gradual and modest pass-through of upstream biofuel price increases to consumers at the pump (Lade and Bushnell, 2019). Figure 11 shows that the biofuel policy shock raises consumer inflation expectations by about 5 basis points. This suggests that persistent price increases in food and energy markets feed into their forward-looking household beliefs about future inflation. Biofuel policy tightening therefore shapes not only realized prices but also inflation expectations.

Figure 10. Impulse responses of energy prices



Notes: The shock is normalized to a 2 percent increase in the real biofuel price. Bands as in Figure 5. Impulse responses are estimated by augmenting the baseline VAR with each biofuel price one at a time. Data are from the U.S. Bureau of Labor Statistics and Bureau of Economic Analysis.

Figure 11. Impulse responses of inflation expectations



Notes: The shock is normalized to a 2 percent increase in the real biofuel price. Bands as in Figure 5. Impulse responses are estimated by augmenting the baseline VAR with inflation expectations. Inflation expectations represent the median expected price change over the next 12 months from the University of Michigan Surveys of Consumers. Data are from the University of Michigan: Inflation Expectation series, retrieved from FRED (accessed at: <https://fred.stlouisfed.org/series/MICH>).

5.5. Distributional consequences

We next examine whether the inflationary effects of biofuel policy shocks differ across income groups. Because food and electricity are basic necessities that account for a larger share of

budgets among lower-income households—and because biofuel policy affects both food prices through feedstock competition and electricity prices through energy cost pass-through—the resulting burden may be unevenly distributed. To address this possibility, we use the distributional CPI (D-CPI) of Jaravel (2026).⁶ The D-CPI follows the construction of the official CPI but applies expenditure weights calculated separately for each income group, allowing inflation exposure to vary with household consumption patterns. As in the heterogeneous price analysis of the previous section, we augment the baseline model with one D-CPI variable at a time. We focus on the lower-middle, middle, and upper-middle income groups, excluding the lowest and highest groups. The lowest income group’s price responses are estimated with considerable uncertainty, reflecting the high variability in their consumption baskets. The highest income group devotes such a small share of its budget to food and energy that the responses are economically negligible.

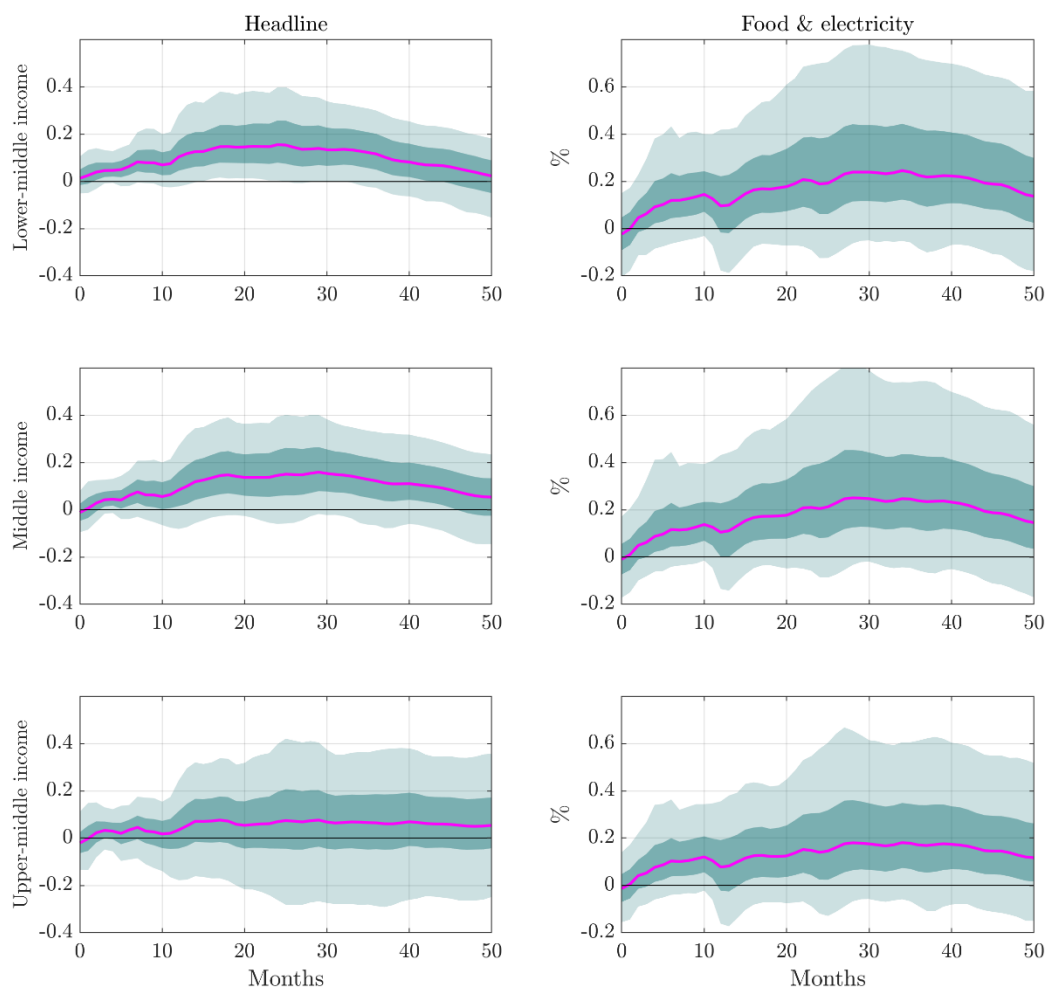
Figure 11 reports the IRFs of the headline D-CPI (left panel) and the food and electricity D-CPI (right panel). The headline responses vary substantially across income groups: the lower-middle-income group shows the strongest reaction, with prices rising significantly, whereas the upper-middle-income group shows a smaller and non-significant response. For food and electricity, responses are more similar but remain larger for lower-middle-income households. At the peak, the index rises by 0.24 percent for the lower-middle-income group, compared with 0.18 percent for the upper-middle-income group, although these effects are statistically significant only at the 68 percent confidence level.

Overall, the estimates suggest that biofuel policy shocks impose a larger inflationary burden on lower-middle-income households, consistent with their greater expenditure shares on food and energy (Sinclair and Stewart, 2026). Because the analysis excludes the lowest income group, these estimates do not directly measure the burden on the poorest households, although their even larger food and energy expenditure shares suggest at least comparable exposure. This

⁶ We use the Chained D-CPI, which combines each group’s price relatives with its realized monthly expenditure shares, thereby allowing for substitution across items as relative prices change (Jaravel, 2026). This approach is consistent with the official Chained CPI methodology. All indices are normalized to 100 in January 1983, and the sample spans January 1983–September 2025. In addition to the headline D-CPI, we construct a food and electricity D-CPI by restricting the expenditure weights to food and electricity categories only, following the same replication code provided by Jaravel (2026). This index measures inflation separately for each income group by constructing group-specific price indices using expenditure weights drawn from the Consumer Expenditure Survey.

burden is likely to grow as U.S. biofuel policy expands—through year-round E15 access, Sustainable Aviation Fuel (SAF) mandates, LCFS expansions, and clean fuel tax credit structures under the Inflation Reduction Act.

Figure 11. Impulse responses of the headline and food and electricity distributional CPIs across income groups



Notes: The shock is normalized to a 2 percent increase in the real biofuel price. Impulse responses are plotted over a fifty-month horizon. Bands as in Figure 5. Each response is estimated by augmenting the baseline VAR with one income-group distributional CPI at a time. The left panels show the headline D-CPI across income groups; the right panels show the food and electricity D-CPI. All series are Chained D-CPIs. The underlying headline D-CPI data are from Xavier Jaravel's webpage (Accessed at: <https://www.xavierjaravel.com/dcpi>); because only the headline series is provided here, we construct the food and electricity D-CPI ourselves using his replication code. Bands as in Figure 5.

6. Robustness checks

This section assesses the sensitivity of the baseline results to instrument predictability, invertibility, lag truncation, and alternative measurement of biofuel prices. The findings remain broadly consistent across specifications.

6.1 Instrument robustness to policy uncertainty

As a robustness check, motivated by mild evidence that the instrument is predictable from policy-uncertainty proxies in the informational sufficiency test, we re-estimate the baseline model using an instrument orthogonalized against three policy-uncertainty indices. The impulse responses are nearly indistinguishable from the baseline; details are in Appendix C.

6.2 Invertibility assumption-Internal instruments approach

The external instruments approach relies on a partial invertibility condition (Stock and Watson, 2018; Miranda-Agrippino and Ricco, 2023), which requires that structural shocks can be expressed as linear combinations of current and lagged reduced-form residuals. To evaluate the sensitivity of our results to this assumption, we re-estimate the model using an internal instruments approach, in which the biofuel policy news series is included directly as the first variable in a recursive VAR, ordered before all other endogenous variables (Plagborg-Møller and Wolf, 2021). The two approaches yield similar directional responses, but the internal instruments estimates are less precise and smaller in magnitude, consistent with measurement error bias that arises when an imperfect proxy is used directly as a structural shock. The greater precision and larger magnitude of the baseline estimates support the use of the external instruments approach as our preferred specification. The impulse responses are provided in Appendix D.

6.3 Lag truncation bias—Local Projections

In finite-order VAR (p) models, impulse response reliability depends on appropriate lag length selection. Misspecification can introduce lag truncation bias, distorting estimated long-run dynamics (see Montiel Olea et al., 2025). The local projections approach avoids this problem by directly estimating impulse responses at each horizon without imposing the cross-horizon dynamic restrictions embedded in a fitted VAR. To evaluate the extent of this potential bias, we re-estimate IRFs using local projections following Jordà (2005):

$$y_{t+h} = \alpha_h + \delta_h u_{1,t} + \theta'_h x_{t-1} + \xi_{h,t} \quad (10)$$

where y_{t+h} denotes the outcome variable at horizon h , $u_{1,t}$ represents the structural biofuel policy shock recovered from the baseline SVAR-IV, and x_{t-1} is a vector of baseline variables. The coefficient δ_h captures the impulse response of y to the biofuel policy shock at horizon h , and $\xi_{h,t}$ is the error term, which may be serially correlated.

Rather than using the raw biofuel news series as the shock, we instead use the structural shock estimated from the baseline VAR to address statistical power concerns in local projections (Nakamura and Steinsson, 2018). The two approaches yield broadly similar results, particularly at short horizons. At longer horizons, local projection estimates are somewhat more volatile and less precise, as each horizon is estimated independently without the smoothing imposed by VAR's dynamic structure. Overall, local projection estimates tend to be larger than the baseline VAR estimates, although the transmission patterns are consistent across both approaches. The results are provided in Appendix D.

6.4 Alternative variable: Replacing biofuel price with ethanol price

In the baseline model, the sample period is limited to February 2025 due to the absence of biodiesel spot price data, which is used as a component of the production-weighted real biofuel price. As a sensitivity check, we replace the real biofuel price with the ethanol nearby futures price, which extends the sample period to December 2025. The results are comparable to the baseline model and somewhat clearer in terms of statistical significance, due to the longer sample period (the impulse responses are shown in Appendix D).

6.5 Different lags

The baseline uses twelve lags, motivated by the annual periodicity of the data-generating process: RFS volume obligations follow an annual rulemaking cycle. Re-estimating with 6, 9, and 18 lags, the first-stage F-statistic falls to 1.82, 3.73, 9.70, against 13.54 in the baseline—only the baseline is strongly identified. The responses share the same sign as the baseline across variables, although the shorter-lag specifications yield attenuated energy and food price responses alongside a larger household food spending response, consistent with the loss of first-stage relevance. The results are provided in Appendix D.

7. Conclusions

This study provides new empirical evidence on how U.S. biofuel policy transmits through agri-food supply chains to retail prices and consumer expenditures. We construct a biofuel policy news series based on regulatory announcements from the EPA and CARB, as well as leaked information from Reuters regarding Renewable Volume Obligations, and use it as an external instrument in a structural VAR framework following the high-frequency identification approach of Känzig (2021). This approach allows us to trace the complete causal pathway from biofuel mandates through commodity markets to retail prices.

We find that tighter biofuel policy generates significant inflationary pressure across both upstream commodity and downstream consumer markets. Field crop prices rise by more than twice the direct feedstock demand effect, and oil prices increase as biofuel compliance costs pass through to petroleum markets. The shock also raises headline consumer prices and food prices, with food-at-home price responses broadly in line with the direct commodity-to-retail pass-through, while household food spending slightly decreases on impact. The inflationary effects are concentrated in food categories most exposed to feedstock competition, particularly fats and oils, meats, poultry and fish, and dairy. Producer food prices respond more strongly than their CPI counterparts, indicating that upstream cost pressures are partially absorbed before reaching consumers. Beyond food markets, retail energy prices rise with a lag, and consumer inflation expectations increase, suggesting that tightening biofuel policy shapes not only realized prices but also forward-looking household beliefs about future inflation.

Historical decompositions show that biofuel policy shocks contributed meaningfully to downstream consumer prices, food-at-home prices, and household food spending. The contribution to food-at-home prices is roughly three times that to food service prices, reflecting the greater commodity intensity of grocery prices. In addition, these shocks account for roughly 1.1 percentage points of the 10.78 percent peak in food price inflation in August 2022. Forecast error variance decompositions further confirm that the contribution of biofuel policy shocks to consumer food prices and food expenditures grows substantially over time, reaching 27-29 percent after three years, consistent with the gradual pass-through of commodity price shocks through supply chains and retail pricing mechanisms. The distributional analysis reveals that the

inflationary consequences of biofuel policy are regressive. Constructing income-group-specific price indices following Jaravel (2026), we show that lower-middle-income households face larger price increases than upper-middle-income households, a pattern consistent with the greater share of food and energy in these households' budgets.

These findings speak directly to ongoing U.S. biofuel policy debates. The recent expansion of year-round E15 access, which allows higher ethanol blends to be sold across more states during summer months, raises feedstock demand in much the same way as the policy stringency shocks we identify. Our results suggest that such expansions, while supporting revenues for field-crop and biofuel producers, generate meaningful inflationary pressure at the retail level, with the heaviest burden falling on lower-income consumers. Similarly, as Low Carbon Fuel Standards spread beyond California to states such as Oregon, Washington, and New Mexico, the compliance cost pressures are likely to grow, with corresponding effects on downstream price transmission.

Several caveats qualify these conclusions. The news series captures policy information conveyed through regulatory announcements and press reports, and surprises transmitted outside this set are not reflected in the instrument. The baseline estimates also rest on a partial invertibility condition, although the internal instruments and local projections exercises in Section 6 deliver consistent transmission patterns. Finally, the distributional analysis relies on income-group expenditure shares rather than household-level price experiences, so the regressivity estimates capture average differences across groups rather than the full dispersion across individual households.

It should be noted that our analysis does not provide a comprehensive welfare evaluation of U.S. biofuel policy. What we do establish is that the costs of these mandates fall systematically on consumers, with the heaviest burden on lower-income households who are least able to absorb them. As policymakers evaluate future Renewable Volume Obligations, LCFS expansions, and clean fuel tax credit structures under the Inflation Reduction Act, a complete accounting of transmission pathways from regulatory announcements through commodity markets to household budgets is necessary for balancing the environmental objectives of renewable fuel policy against its economic and equity consequences.

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Online Appendix

Appendix A: U.S. biofuel policy announcements

This appendix documents the biofuel policy announcements used to construct the biofuel policy news series. Table A.1 lists all announcements by date, announcement title, and source. The final sample comprises 98 announcements from November 2007 to February 2025: 29 from EPA, 53 from CARB, and 16 from Reuters. EPA announcements include proposed and finalized Renewable Volume Obligations (RVOs) for the Renewable Fuel Standard (RFS) for the following compliance year. CARB announcements consist of quarterly Low Carbon Fuel Standard (LCFS) compliance reports, which document total credits and deficits generated by alternative fuels including biodiesel, renewable diesel, ethanol, biomethane, fossil natural gas, and electricity. Reuters articles include reports on leaked EPA regulatory decisions and exclusive reporting on anticipated RFS policy actions.

Table A.1. U.S. biofuel policy announcements, 11/2007-2/2025

Date	Announcements	Source
11/27/2007	EPA published a Notice in the Federal Register announcing a renewable fuel standard for 2008 of 4.66%	EPA
2/7/2008	EPA Proposes Regulations for 2007 and Beyond Renewable Fuel Standard (RFS) Program	Reuters
11/17/2008	Renewable Fuel Standard Increased for 2009	EPA
5/5/2009	EPA Lays out a Plan for the Nation's Increase in Renewable fuels	EPA
3/26/2010	Regulation of Fuels and Fuel Additives: Changes to Renewable Fuel Standard Program (Final rule)	EPA
7/12/2010	EPA Proposes 2011 Renewable Fuel Standards	EPA
11/29/2010	EPA Finalizes 2011 Renewable Fuel Standards	EPA
6/21/2011	EPA Proposes 2012 Renewable Fuel Standards, 2013 Biomass-Based Diesel Volume	EPA
8/22/2011	Low Carbon Fuel Standard (LCFS) quarterly reports	CARB
10/27/2011	Low Carbon Fuel Standard (LCFS) quarterly reports	CARB
12/27/2011	EPA Finalizes 2012 Renewable Fuel Standards	EPA
2/2/2012	Southwest Iowa Renewable Energy, LLC to Pay \$10,150 for Risk Management Plan Violations at Council Bluffs Ethanol Plant	EPA
2/17/2012	Low Carbon Fuel Standard (LCFS) quarterly reports	CARB
4/2/2012	EPA to Allow 15 Percent Renewable Fuel in Gasoline/Agency approves first applications for registration of ethanol to make E15	EPA
6/22/2012	Low Carbon Fuel Standard (LCFS) quarterly reports	CARB
10/31/2012	Low Carbon Fuel Standard (LCFS) quarterly reports	CARB
12/31/2012	Low Carbon Fuel Standard (LCFS) quarterly reports	CARB

1/31/2013	EPA Proposes 2013 Renewable Fuel Standards	EPA
3/29/2013	Low Carbon Fuel Standard (LCFS) quarterly reports	CARB
7/31/2013	Low Carbon Fuel Standard (LCFS) quarterly reports	CARB
8/6/2013	EPA Finalizes 2013 Renewable Fuel Standards to Help Promote American Energy Independence, Reduce Carbon Pollution/EPA also announces steps to address concerns about the E10 blend wall	EPA
9/30/2013	Low Carbon Fuel Standard (LCFS) quarterly reports	CARB
11/15/2013	EPA Proposes 2014 Renewable Fuel Standards / Proposal Seeks Input to Address "E10 Blend Wall," Reaffirms Commitment to Biofuels	EPA
1/27/2014	Low Carbon Fuel Standard (LCFS) quarterly reports	CARB
4/7/2014	Low Carbon Fuel Standard (LCFS) quarterly reports	CARB
7/7/2014	Low Carbon Fuel Standard (LCFS) quarterly reports	CARB
10/17/2014	Low Carbon Fuel Standard (LCFS) quarterly reports	CARB
1/6/2015	Low Carbon Fuel Standard (LCFS) quarterly reports	CARB
3/16/2015	Low Carbon Fuel Standard (LCFS) quarterly reports	CARB
5/29/2015	EPA Proposes Increases in Renewable Fuel Levels	EPA
6/30/2015	Low Carbon Fuel Standard (LCFS) quarterly reports	CARB
9/23/2015	Low Carbon Fuel Standard (LCFS) quarterly reports	CARB
11/30/2015	EPA Finalizes Increases in Renewable Fuel Levels	EPA
12/11/2015	Low Carbon Fuel Standard (LCFS) quarterly reports	CARB
4/15/2016	Low Carbon Fuel Standard (LCFS) quarterly reports	CARB
5/18/2016	EPA Proposes Increase in Renewable Fuel Levels	EPA
7/28/2016	Low Carbon Fuel Standard (LCFS) quarterly reports	CARB
10/20/2016	Low Carbon Fuel Standard (LCFS) quarterly reports	CARB
11/23/2016	EPA Finalizes Increase in Renewable Fuel Volumes	EPA
1/23/2017	Low Carbon Fuel Standard (LCFS) quarterly reports	CARB
4/19/2017	Low Carbon Fuel Standard (LCFS) quarterly reports	CARB
7/5/2017	EPA Proposes RFS Volumes Reflective of Market Realities for 2018	EPA
8/2/2017	Low Carbon Fuel Standard (LCFS) quarterly reports	CARB
10/31/2017	Low Carbon Fuel Standard (LCFS) quarterly reports	CARB
11/30/2017	U.S. EPA ups biofuel targets slightly, draws scorn from refiners	Reuters
1/31/2018	Low Carbon Fuel Standard (LCFS) quarterly reports	CARB
4/25/2018	Low Carbon Fuel Standard (LCFS) quarterly reports	CARB
6/26/2018	Proposed Volume Standards for 2019 and the Biomass-Based Diesel Volume for 2020	EPA
7/11/2018	EPA scraps detailed plan to force U.S. refiners to blend more biofuels	Reuters
7/31/2018	Low Carbon Fuel Standard (LCFS) quarterly reports	CARB
9/20/2018	EPA details broad expansion of biofuel waiver program	Reuters
10/31/2018	Low Carbon Fuel Standard (LCFS) quarterly reports	CARB
11/30/2018	EPA Finalizes RFS Volumes for 2019 and Biomass Based Diesel Volumes for 2020	EPA
1/31/2019	Low Carbon Fuel Standard (LCFS) quarterly reports	CARB
3/12/2019	EPA Proposes Rule to Allow E15 Waiver and to Improve RIN Market Transparency	EPA
4/30/2019	Low Carbon Fuel Standard (LCFS) quarterly reports	CARB
5/31/2019	EPA Delivers on President Trump's Promise to Allow Year-Round Sale of E15 Gasoline and Improve Transparency in Renewable Fuel Markets	EPA

7/5/2019	EPA Proposes Renewable Volume Obligations for 2020 On Track to Meet Statutory Deadline	EPA
7/31/2019	Low Carbon Fuel Standard (LCFS) quarterly reports	CARB
10/31/2019	Low Carbon Fuel Standard (LCFS) quarterly reports	CARB
12/19/2019	EPA Fulfills Another Trump Administration Promise: Finalizes RFS Volumes for 2020 and Biomass Based Diesel Volumes for 2021	EPA
1/31/2020	Low Carbon Fuel Standard (LCFS) quarterly reports	CARB
4/30/2020	Low Carbon Fuel Standard (LCFS) quarterly reports	CARB
7/31/2020	Low Carbon Fuel Standard (LCFS) quarterly reports	CARB
9/9/2020	Exclusive: Trump administration to deny pending retroactive U.S. biofuel waivers — sources	Reuters
10/30/2020	Low Carbon Fuel Standard (LCFS) quarterly reports	CARB
1/11/2021	EXCLUSIVE Trump administration expected to grant biofuel waivers for some oil refiners - sources	Reuters
1/15/2021	Notice of Proposed Rulemaking for E15 Labeling	EPA
1/29/2021	Low Carbon Fuel Standard (LCFS) quarterly reports	CARB
4/30/2021	Low Carbon Fuel Standard (LCFS) quarterly reports	CARB
7/30/2021	Low Carbon Fuel Standard (LCFS) quarterly reports	CARB
9/22/2021	EXCLUSIVE: Biden administration mulls big cuts to biofuel mandates in win for oil industry — document	Reuters
10/29/2021	Low Carbon Fuel Standard (LCFS) quarterly reports	CARB
12/7/2021	Proposed Volume Standards for 2020, 2021, and 2022	EPA
1/12/2022	Exclusive: Biden weighing cuts to 2022 ethanol blending mandate proposal	Reuters
1/31/2022	Low Carbon Fuel Standard (LCFS) quarterly reports	CARB
4/29/2022	Low Carbon Fuel Standard (LCFS) quarterly reports	CARB
6/1/2022	EXCLUSIVE: Biden administration likely to retroactively raise ethanol blending volumes for 2021, sources say	Reuters
7/29/2022	Low Carbon Fuel Standard (LCFS) quarterly reports	CARB
10/31/2022	Low Carbon Fuel Standard (LCFS) quarterly reports	CARB
12/1/2022	EPA Takes Next Steps in Renewable Fuel Standard Program for 2023–25	EPA
1/31/2023	Low Carbon Fuel Standard (LCFS) quarterly reports	CARB
3/1/2023	EPA proposes expanded sales of higher ethanol blend gasoline in US Midwest	Reuters
4/19/2023	Biden administration sees year-round sales of higher ethanol fuel by 2024	Reuters
4/28/2023	Low Carbon Fuel Standard (LCFS) quarterly reports	CARB
6/6/2023	EPA to remove proposed EV volumes from biofuel blending rule	Reuters
6/21/2023	Final Renewable Fuels Standards for 2023, 2024, and 2025	EPA
7/31/2023	Low Carbon Fuel Standard (LCFS) quarterly reports	CARB
10/31/2023	Low Carbon Fuel Standard (LCFS) quarterly reports	CARB
1/31/2024	Low Carbon Fuel Standard (LCFS) quarterly reports	CARB
2/20/2024	Exclusive: Biden administration to approve E15 gasoline expansion starting in 2025, sources say	Reuters
4/5/2024	Exclusive: Biden administration's initial SAF subsidy model to raise climate hurdle for ethanol	Reuters
4/18/2024	Exclusive: US EPA to announce temporary expanded sales of higher-ethanol gasoline blend — sources	Reuters
4/30/2024	Low Carbon Fuel Standard (LCFS) quarterly reports	CARB
7/31/2024	Low Carbon Fuel Standard (LCFS) quarterly reports	CARB

10/31/2024	Low Carbon Fuel Standard (LCFS) quarterly reports	CARB
1/10/2025	Exclusive: US to release clean fuel tax credit model that limits ethanol producers' credit access, sources say	Reuters
2/21/2025	Ahead of Summer Driving Season, EPA Allows Expanded E15 Access to Midwest States Year-Round	EPA

Notes: The table lists all 98 biofuel policy announcements used to construct the biofuel policy news series, spanning November 2007 to February 2025. EPA = Environmental Protection Agency; CARB = California Air Resources Board. Reuters announcements report on or anticipate EPA Renewable Volume Obligation decisions. CARB announcements are quarterly compliance reports on Low Carbon Fuel Standard (LCFS) credits and deficits.

Appendix B: Data description and source

Table B. Data description and source for selected variables

Variable (Unit)	Description	Source
A. Instrument		
Biofuel policy news (%)	First principal component of daily changes in ethanol futures prices (nearby and first-deferred contract maturities) around regulatory announcements from EPA, CARB, and Reuters, weighted by production shares	Bloomberg terminal, EIA, and authors' calculations
B. Baseline		
Real biofuel price	Production-weighted composite of the Chicago ethanol (Platts) nearby futures contract price and the biodiesel spot price, weighted by monthly production shares and deflated by CPI	Bloomberg terminal; USDA bioenergy Statistics
Real oil price (index, 2017=100)	Global Brent crude spot price, deflated by CPI	FRED; BLS
Real field crop price (index, 2017=100)	Production-weighted average of nearby corn and soybean futures prices, deflated by CPI	Bloomberg terminal; USDA NASS; BLS
CPI (index, 1982-84=100)	Headline Consumer Price Index for all urban consumers (CPI-U), all items	BLS
RPCE for food (Billions of Chained 2017 Dollars)	Real Personal Consumption Expenditures price: food	BEA
Food-at-home CPI (index, 1982-84=100)	Consumer Price Index for food at home in U.S. city average (grocery store prices)	BLS
Food-away-from-home CPI (index, 1982-84=100)	Consumer Price Index for food away from home in U.S. city average (restaurant prices)	BLS
PCE gasoline (index, 2017=100)	Personal Consumption Expenditures price index for gasoline and other energy goods	BEA

C. Wider Effects

Consumer food price indices (CPI)

Fats and oils CPI (index, 1982-84=100)	Consumer Price Index for fats and oils in U.S. city average	BLS
Cereals and bakery CPI (index, 1982-84=100)	Consumer Price Index for cereals and bakery products in U.S. city average	BLS
Meats, poultry, fish, and eggs CPI (index, 1982-84=100)	Consumer Price Index for meats, poultry, fish, and eggs in U.S. city average	BLS
Dairy CPI (index, 1982-84=100)	Consumer Price Index for dairy and related products in U.S. city average	BLS
Fruits and vegetables CPI (index, 1982-84=100)	Consumer Price Index for fruits and vegetables in U.S. city average	BLS

Producer food price indices (PPI)

Fats and oils PPI (index, 1981=100)	Producer price index for fats, oils refining and blending: shortening and cooking oils	BLS
Cereal and bakery PPI (index, 1982=100)	Producer price index for cereal and bakery products	BLS
Meats, poultry, fish, and egg PPI (index, 1982=100)	Producer price index for meats, poultry, and fish	BLS
Dairy PPI (index, 1982=100)	Producer price index for dairy products	BLS
Fruits and vegetables PPI (index, 1982=100)	Producer price index for processed fruits and vegetables	BLS

Energy prices

CPI fuel and oil (index, 1982-84=100)	Consumer Price Index for fuel oil and other fuels	BLS
PCE gasoline (index, 2017=100)	Personal Consumption Expenditures price index for gasoline and other energy goods	BEA
Consumer inflation expectations (%)	University of Michigan Survey of Consumers: median expected price change over the next 12 months	University of Michigan

D. Distributional Effects

Headline distributional CPI

Headline D-CPI, lower-middle income (index)	Distributional headline CPI (all items) for the lower-middle income group (second income quintile), constructed following Jaravel (2026) by reweighting BLS item-strata price relatives with group-specific expenditure shares from the CEX	BLS; CEX; authors' calculations
Headline D-CPI, middle income (index)	Distributional headline CPI (all items) for the middle income group (third income quintile), constructed as above	BLS; CEX; authors' calculations

Headline D-CPI, upper-middle income (index)	Distributional headline CPI (all items) for the upper-middle income group (fourth income quintile), constructed as above	BLS; CEX; authors' calculations
<i>Food and electricity distributional CPI</i>		
Food and electricity D-CPI, lower-middle income (index)	Distributional CPI covering food (58 food-at-home and food-away-from-home item strata) and electricity (stratum SEHF01) for the lower-middle income group (second income quintile), constructed following Jaravel (2026) by reweighting BLS strata price relatives with group-specific CEX expenditure shares	BLS; CEX; authors' calculations
Food and electricity D-CPI, middle income (index)	Food and electricity distributional CPI for the middle income group (third income quintile), constructed as above	BLS; CEX; authors' calculations
Food and electricity D-CPI, upper-middle income (index)	Food and electricity distributional CPI for the upper-middle income group (fourth income quintile), constructed as above	BLS; CEX; authors' calculations
E. Other Variables		
Exchange rate (index, 2020=100)	U.S. real narrow effective exchange rate	BIS
Ethanol production (Million Gallons)	U.S. fuel ethanol production	EIA
Biodiesel production (Million Gallons)	U.S. biodiesel production	EIA
Real ethanol price	Chicago ethanol (Platts) nearby futures price, deflated by CPI	Bloomberg terminal

Notes: All price variables are transformed to natural logarithms in the empirical analysis. Ethanol, oil, and field crop prices are deflated by the headline CPI. Sample period is November 2007 to February 2025. BLS = Bureau of Labor Statistics; BEA = Bureau of Economic Analysis; FRED = Federal Reserve Economic Data; CEX = Consumer Expenditure Survey; CARB = California Air Resources Board; EPA = Environmental Protection Agency; USDA = U.S. Department of Agriculture; BIS = Bank for International Settlements.

Appendix C: Informational-sufficiency tests and policy-uncertainty-orthogonalized instrument

C.1 Informational-sufficiency test results

In this section, we report the full results of the informational-sufficiency test introduced in Section 3.4. The proxy-SVAR identification requires that the instrument be correlated with the biofuel policy shock but uncorrelated with other structural shocks. A necessary condition, formalized by Mori and Peersman (2026), is that the instrument not be predictable from information available to agents at the time—if lagged macroeconomic or financial variables forecast the identified shock, the instrument is capturing anticipated or confounding movements rather than genuine

policy news, and the exclusion restriction fails. The test therefore checks whether lagged information predicts the instrument (and the shock it identifies) after conditioning on the information already in the VAR. Tables C.1 and C.2 present Granger-predictability p -values by information block and by individual series, respectively, for the full sample and for a pre-COVID subsample ending in December 2019. Each cell tests whether twelve monthly lags of the relevant information predict one of three outcome objects—the raw news series, the reduced-form residual of the real biofuel price equation, or the identified structural shock. The p (*orth.*) column, which orthogonalizes the outcome on the baseline VAR variables, carries the verdict; large values indicate sufficiency.

Over the full sample the instrument passes cleanly. Table C.1 shows that no information block predicts the structural shock: the orthogonalized p -values are 0.76, 0.58, 1.00, 0.83, and 0.51 for the financial, policy-uncertainty, commodity, monetary, and weather blocks, respectively (Table C.1, Panel C), and the reduced-form residual gives the same conclusion (Panel B). For the surprise proxy (Panel A), the monetary policy and weather appear to predict the raw news series, but neither survives conditioning on the baseline VAR lags. In Table C.2, the only orthogonalized rejection for the structural shock is a single financial subindex (nonfinancial leverage, $p = 0.074$, Panel C), and for the news series the severe weather index is marginally significant ($p = 0.058$, Panel A).

The pre-COVID subsample reveals one sensitivity. In Table C.1, the policy-uncertainty block is marginally predictive of the structural shock (joint orthogonalized $p = 0.022$, Panel C); no other block rejects. Table C.2 locates this in the economic policy uncertainty index ($p = 0.041$, Panel C) and, more weakly, the geopolitical risk index ($p = 0.081$, Panel C); the federal funds rate is also borderline ($p = 0.097$, Panel C), although it does not reject as part of the monetary block. A few series are borderline for the auxiliary objects as well—the supply-chain pressure index for the reduced-form residual ($p = 0.044$, Panel B) and the severe-weather index for the news series ($p = 0.058$, Panel A). At the block level, Because the sensitivity is concentrated in the policy uncertainty block, we address it directly through the orthogonalized-instrument check in Section C.2.

Table C.1. Informational-sufficiency tests for the biofuel-policy instrument (block-joint)

Information block	Full sample		Pre-COVID ($\leq 2019M12$)	
	<i>p (raw)</i>	<i>p (orth.)</i>	<i>p (raw)</i>	<i>p (orth.)</i>
Panel A. Instrument (surprise proxy)				
Financial / credit	0.908	0.983	—	—
Policy uncertainty	0.982	1.000	0.821	0.678
Commodity / global	0.360	0.954	0.681	0.910
Monetary policy	0.005	0.338	1.000	0.992
Weather	0.052	0.232	0.052	0.232
Panel B. Reduced-form residual				
Financial / credit	0.236	0.268	—	—
Policy uncertainty	0.774	0.787	0.179	0.166
Commodity / global	0.841	0.827	0.352	0.373
Monetary policy	0.950	0.948	0.756	0.749
Weather	0.439	0.444	0.439	0.444
Panel C. Structural shock				
Financial / credit	0.745	0.759	—	—
Policy uncertainty	0.557	0.575	0.022	0.022
Commodity / global	0.999	0.999	0.921	0.926
Monetary policy	0.842	0.829	0.497	0.488
Weather	0.517	0.512	0.517	0.512

Notes: Each cell reports the *p*-value of a Granger F-test that twelve monthly lags of the information block jointly predict the outcome object, controlling for a constant and twelve own lags. *p (raw)* uses the outcome as measured; *p (orth.)* uses the outcome after orthogonalizing it on 12 lags of the baseline VAR variables (the Mori-Peersman informational-sufficiency criterion). Small *p*-values indicate predictability, i.e., a potential sufficiency violation; large *p (orth.)* values indicate the instrument is sufficient against that block. Em-dashes denote tests not computable on the pre-COVID subsample (the nine-series financial block exhausts the available degrees of freedom at 12 lags; see the per-series table). N ranges 168-184 (full) and equals 122 (pre-COVID), except the weather block (N = 116), reflecting series coverage. Blocks group the series listed in Table C.2: financial/credit (nine series), policy uncertainty (GPR, TPU, EPU), commodity/global (GSCPI and oil supply news), monetary policy (FFR and the BRW and BS shocks), and weather (KMP severe-weather and agroclimatic indices). Series definitions and sources are given in the notes to Table C.2.

Table C.2. Informational-sufficiency tests for the biofuel-policy instrument (per series)

Series	Full sample		Pre-COVID ($\leq 2019M12$)	
	<i>p (raw)</i>	<i>p (orth.)</i>	<i>p (raw)</i>	<i>p (orth.)</i>
Panel A. Instrument (surprise proxy)				
<i>Financial / credit</i>				
Financial uncertainty (LMN)	0.711	1.000	0.905	0.983
Excess bond premium	0.826	0.975	0.919	0.796
Gilchrist–Zakrajšek spread	0.806	1.000	1.000	0.975
NFCI	0.461	0.909	0.943	0.976
Adjusted NFCI	0.301	0.956	0.901	0.919
NFCI: risk	0.406	0.917	0.949	0.986

Series	Full sample		Pre-COVID ($\leq 2019M12$)	
	<i>p (raw)</i>	<i>p (orth.)</i>	<i>p (raw)</i>	<i>p (orth.)</i>
NFCI: credit	0.628	0.972	0.989	0.976
NFCI: leverage	0.596	0.940	0.282	0.301
Nonfinancial leverage	0.371	0.992	0.760	0.968
<i>Policy uncertainty</i>				
Geopolitical risk (GPR)	0.833	0.988	0.709	0.712
Trade policy uncertainty (TPU)	0.675	1.000	0.919	0.978
Economic policy uncertainty (EPU)	0.639	0.942	0.662	0.287
<i>Commodity / global</i>				
Supply-chain pressure (GSCPI)	0.109	0.723	0.292	0.631
Oil supply news	0.502	0.909	0.944	0.867
<i>Monetary policy</i>				
Federal funds rate	0.060	0.682	0.981	0.946
Monetary policy shock (BRW)	0.503	0.834	0.909	0.818
Monetary policy shock (BS)	0.004	0.839	0.899	0.968
<i>Weather</i>				
Severe weather (KMP)	0.031	0.058	0.031	0.058
Agroclimatic index (KMP)	0.542	0.223	0.542	0.223

Panel B. Reduced-form residual

Financial / credit

Financial uncertainty (LMN)	0.635	0.699	0.272	0.279
Excess bond premium	0.640	0.651	0.149	0.163
Gilchrist–Zakrajšek spread	0.994	0.994	0.373	0.389
NFCI	0.988	0.989	0.574	0.590
Adjusted NFCI	0.987	0.987	0.552	0.576
NFCI: risk	0.994	0.994	0.722	0.740
NFCI: credit	0.928	0.939	0.422	0.450
NFCI: leverage	0.732	0.746	0.495	0.497
Nonfinancial leverage	0.781	0.815	0.788	0.800

Policy uncertainty

Geopolitical risk (GPR)	0.254	0.260	0.069	0.070
Trade policy uncertainty (TPU)	0.987	0.989	0.994	0.992
Economic policy uncertainty (EPU)	0.657	0.708	0.359	0.362

Commodity / global

Supply-chain pressure (GSCPI)	0.497	0.494	0.040	0.044
Oil supply news	0.926	0.913	0.965	0.966

Monetary policy

Federal funds rate	0.998	0.998	0.750	0.735
Monetary policy shock (BRW)	0.825	0.806	0.758	0.756
Monetary policy shock (BS)	0.451	0.479	0.162	0.179

Weather

Severe weather (KMP)	0.371	0.386	0.371	0.386
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Series	Full sample		Pre-COVID ($\leq 2019M12$)	
	<i>p (raw)</i>	<i>p (orth.)</i>	<i>p (raw)</i>	<i>p (orth.)</i>
Agroclimatic index (KMP)	0.604	0.593	0.604	0.593
Panel C. Structural shock				
<i>Financial / credit</i>				
Financial uncertainty (LMN)	0.259	0.285	0.521	0.531
Excess bond premium	0.994	0.994	0.928	0.925
Gilchrist–Zakrajšek spread	0.883	0.884	0.381	0.394
NFCI	0.958	0.960	0.960	0.962
Adjusted NFCI	0.981	0.982	0.921	0.922
NFCI: risk	0.978	0.980	0.953	0.956
NFCI: credit	0.812	0.823	0.849	0.855
NFCI: leverage	0.625	0.625	0.701	0.705
Nonfinancial leverage	0.065	0.074	0.659	0.673
<i>Policy uncertainty</i>				
Geopolitical risk (GPR)	0.787	0.784	0.078	0.081
Trade policy uncertainty (TPU)	0.958	0.966	0.284	0.291
Economic policy uncertainty (EPU)	0.156	0.176	0.043	0.041
<i>Commodity / global</i>				
Supply-chain pressure (GSCPI)	0.941	0.944	0.694	0.700
Oil supply news	0.997	0.997	0.969	0.970
<i>Monetary policy</i>				
Federal funds rate	0.994	0.993	0.100	0.097
Monetary policy shock (BRW)	0.348	0.328	0.515	0.499
Monetary policy shock (BS)	0.685	0.697	0.132	0.145
<i>Weather</i>				
Severe weather (KMP)	0.718	0.731	0.718	0.731
Agroclimatic index (KMP)	0.530	0.523	0.530	0.523

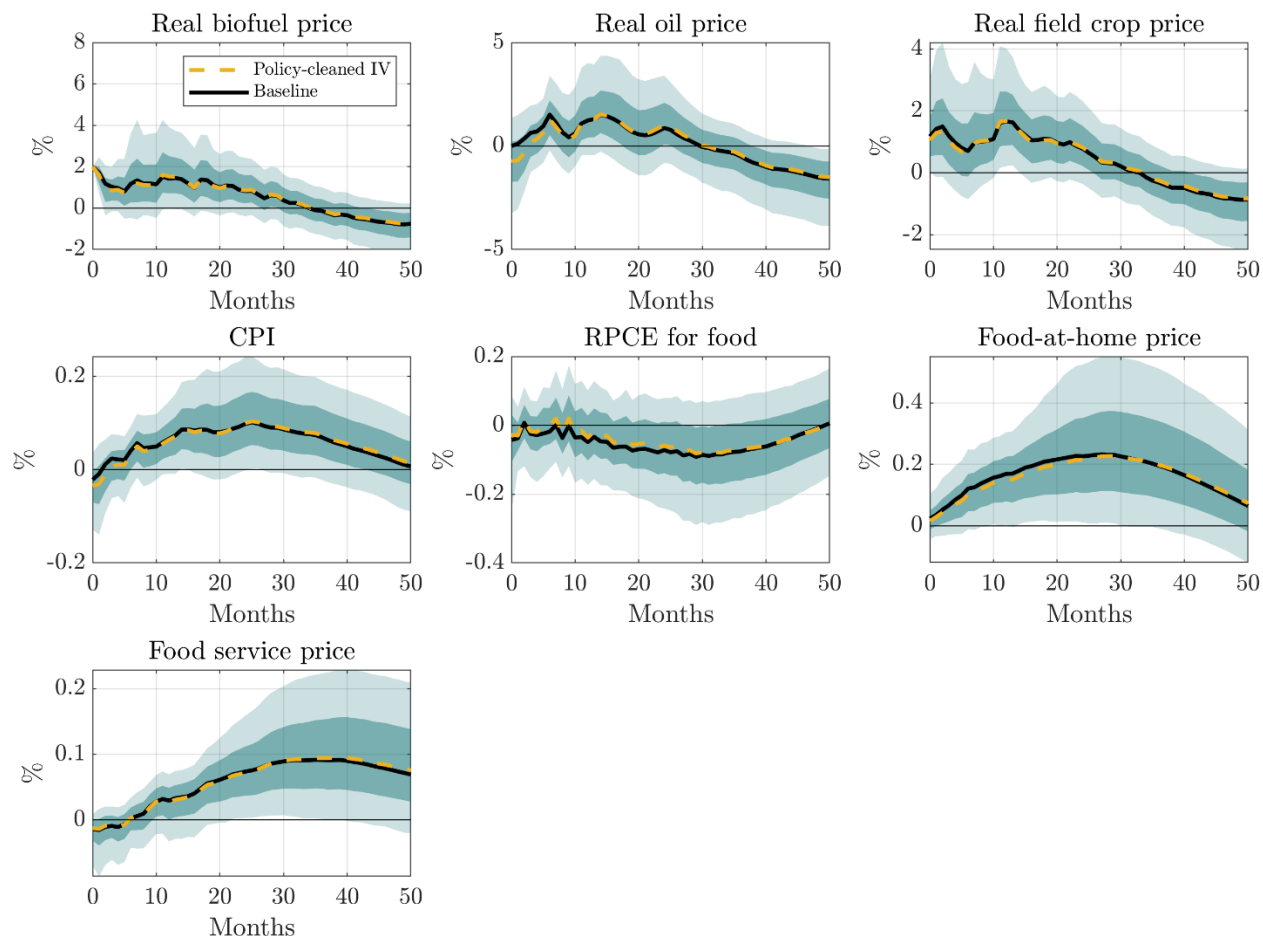
Series definitions and sources: Financial / credit: financial uncertainty (LMN), the one-month-ahead financial-uncertainty index of Ludvigson, Ma, and Ng (2021); excess bond premium (EBP) and the Gilchrist–Zakrajšek (GZ) spread, Gilchrist and Zakrajšek (2012), updated by the Federal Reserve Board; the National Financial Conditions Index (NFCI), the Adjusted NFCI (ANFCI), and its risk, credit, leverage, and nonfinancial-leverage subindexes, Federal Reserve Bank of Chicago. Policy uncertainty: geopolitical risk (GPR), Caldara and Iacoviello (2022); trade policy uncertainty (TPU) and economic policy uncertainty (EPU), Baker, Bloom, and Davis (2016). Commodity / global: Global Supply Chain Pressure Index (GSCPI), Federal Reserve Bank of New York; oil supply news, Känzig (2021). Monetary policy: effective federal funds rate (FFR), Board of Governors of the Federal Reserve System; monetary policy shock (BRW), Bu, Rogers, and Wu (2021); monetary policy shock (BS), Bauer and Swanson (2023). Weather: severe-weather and agroclimatic index (ACI) series (KMP), Kim, Matthes, and Phan (2025).

Notes: Per-series counterpart of Table C.1; each row tests a single series (constant, twelve own lags of the outcome, and twelve lags of the series). *p (orth.)* orthogonalizes the outcome on twelve lags of the baseline VAR variables; small values indicate predictability. Panels report the three outcome objects: the news series (Panel A), the reduced-form residual (Panel B), and the structural shock (Panel C). N = 184 (full) / 122 (pre-COVID) except BRW (168), BS (171 full), and the weather series (116); the KMP weather and agroclimatic series end May 2019, so their full and pre-COVID entries coincide.

C.2 Policy-uncertainty-orthogonalized instrument

Section C.1 shows that, although the biofuel policy news series is informationally sufficient over the full sample, policy uncertainty is marginally predictive of the structural shock in the pre-2020 subsample. This raises the possibility that part of the series reflects anticipated policy conditions rather than genuine news. If identification rested on that forecastable component, the estimated responses would capture the effects of expected policy conditions rather than the unanticipated policy surprises the instrument is designed to isolate. As a robustness check, we orthogonalize the instrument against twelve lags of three policy-uncertainty proxies (the geopolitical risk index, trade policy uncertainty, and economic policy uncertainty) and retain the residual, rescaled to the mean and standard deviation of the original series. Figure C.1 compares the impulse responses to the baseline. The impulse responses with the cleaned instrument are comparable to the baseline, matching it in direction and closely in magnitude, though somewhat less precisely estimated.

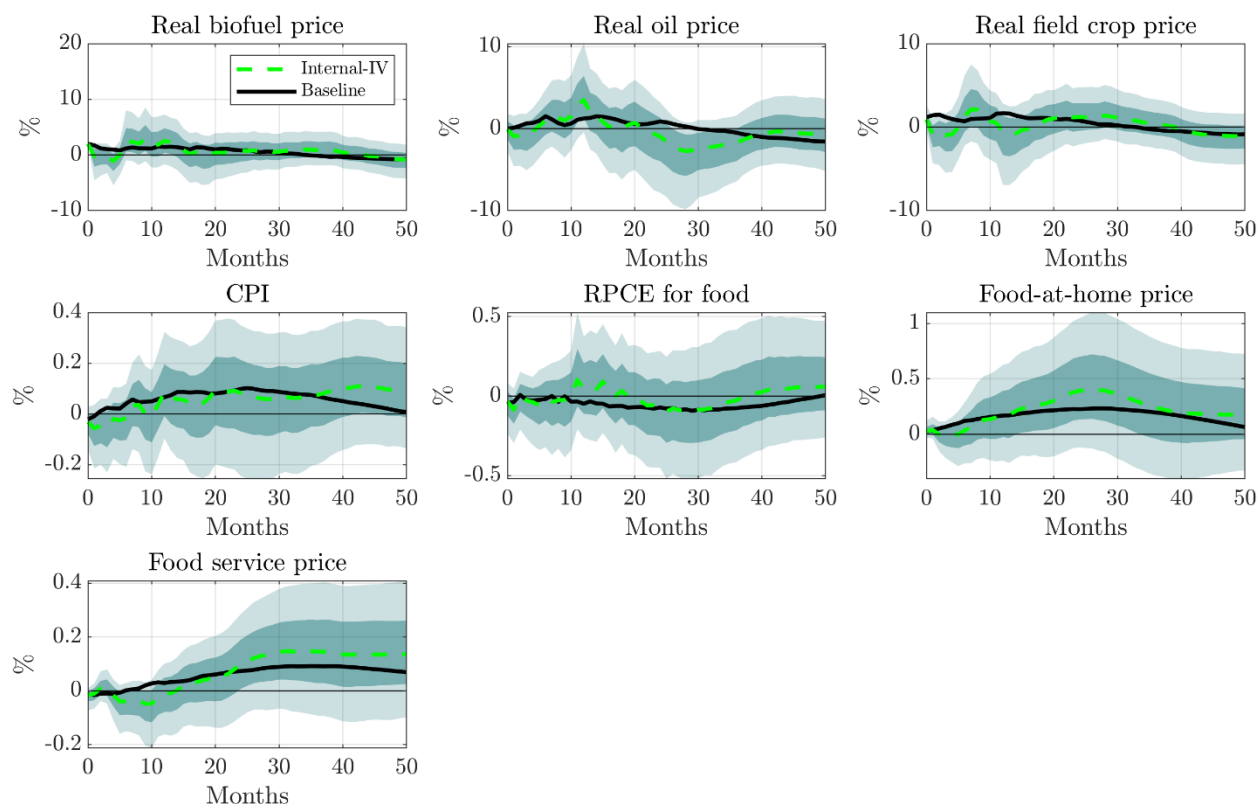
Figure C.1. Impulse responses to a biofuel policy shock: policy-cleaned instrument versus baseline



Notes: The figure compares impulse responses identified with the policy-cleaned instrument (dashed line) to the baseline responses estimated original instrument (solid black line). The shock is normalized to a two percent increase in the real biofuel price on impact. Horizons are in months. The policy-cleaned instrument is the residual from a regression of the original instrument on twelve lags of the geopolitical risk index, trade policy uncertainty, and economic policy uncertainty, rescaled to the original mean and standard deviation; its heteroskedasticity-robust first stage F-statistic is 14.7. Bands as in Figure 5.

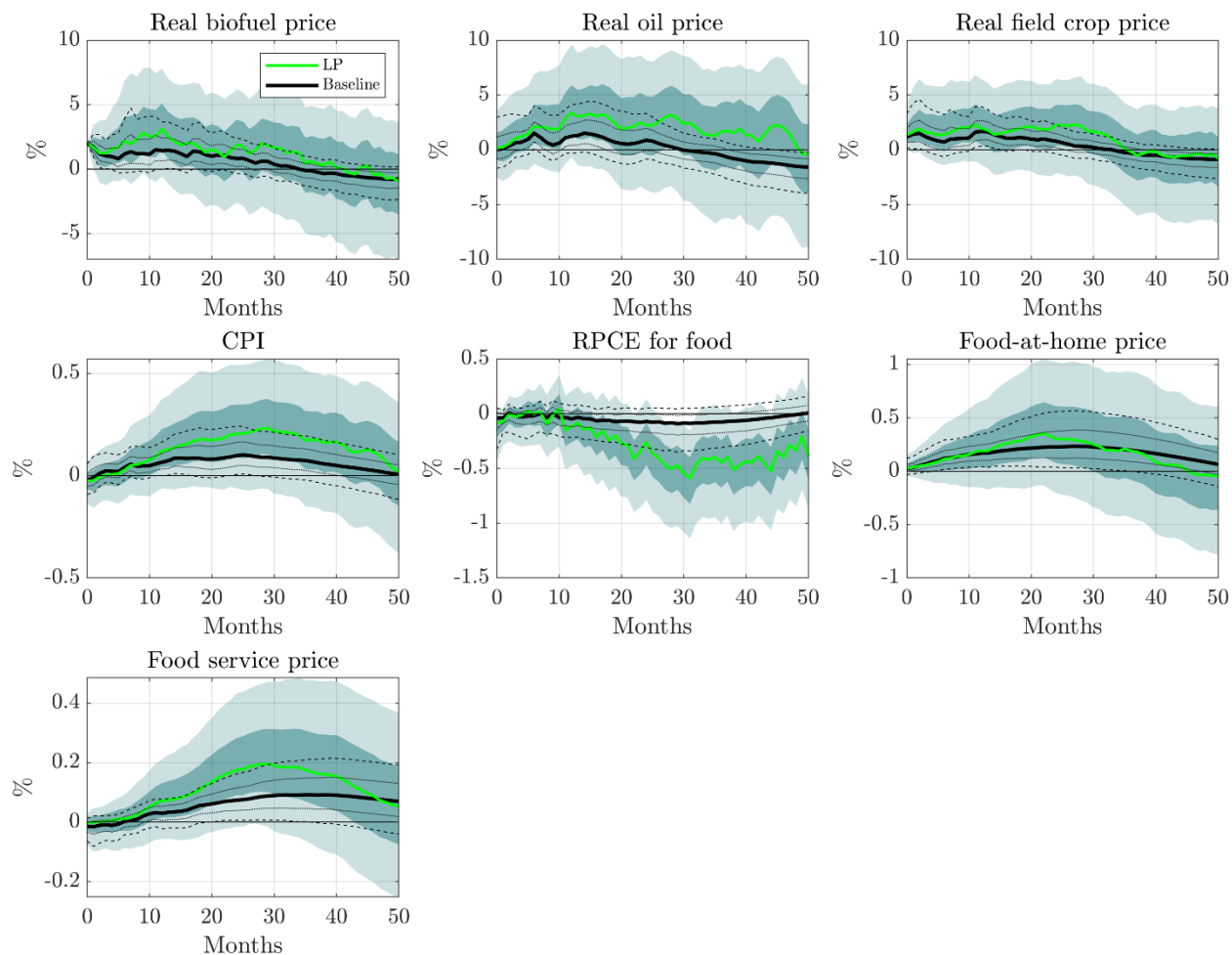
Appendix D: Sensitivity and robustness checks

Figure D.1. Impulse responses to a biofuel policy shock using internal instruments approach



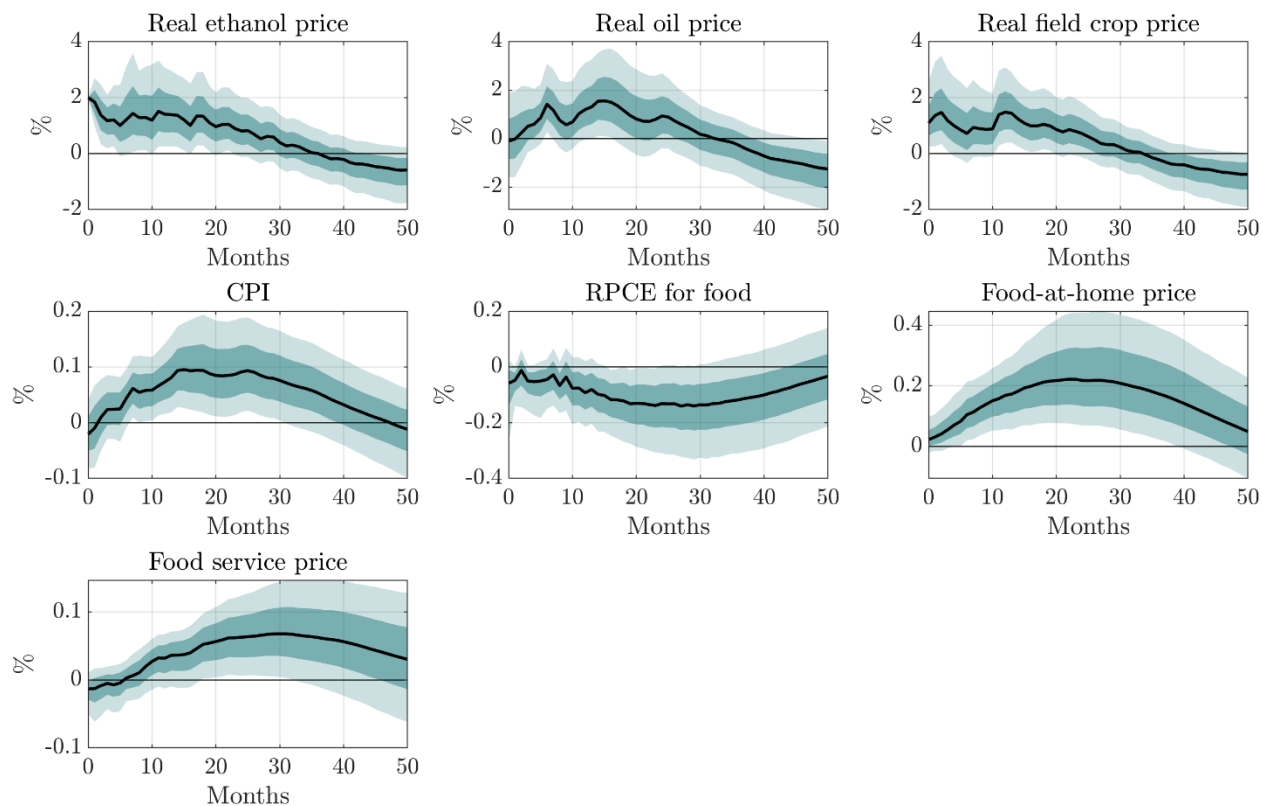
Notes: Impulse responses are estimated using both the internal instruments approach and the external instruments (SVAR-IV) approach. Responses are normalized to raise real ethanol price by 2 percent. The green dashed line represents the point estimate from the internal instruments approach, and the black solid line represents the estimate from the external instruments approach. Bands as in Figure 5.

Figure D.2. Impulse responses to a biofuel policy shock using local projections



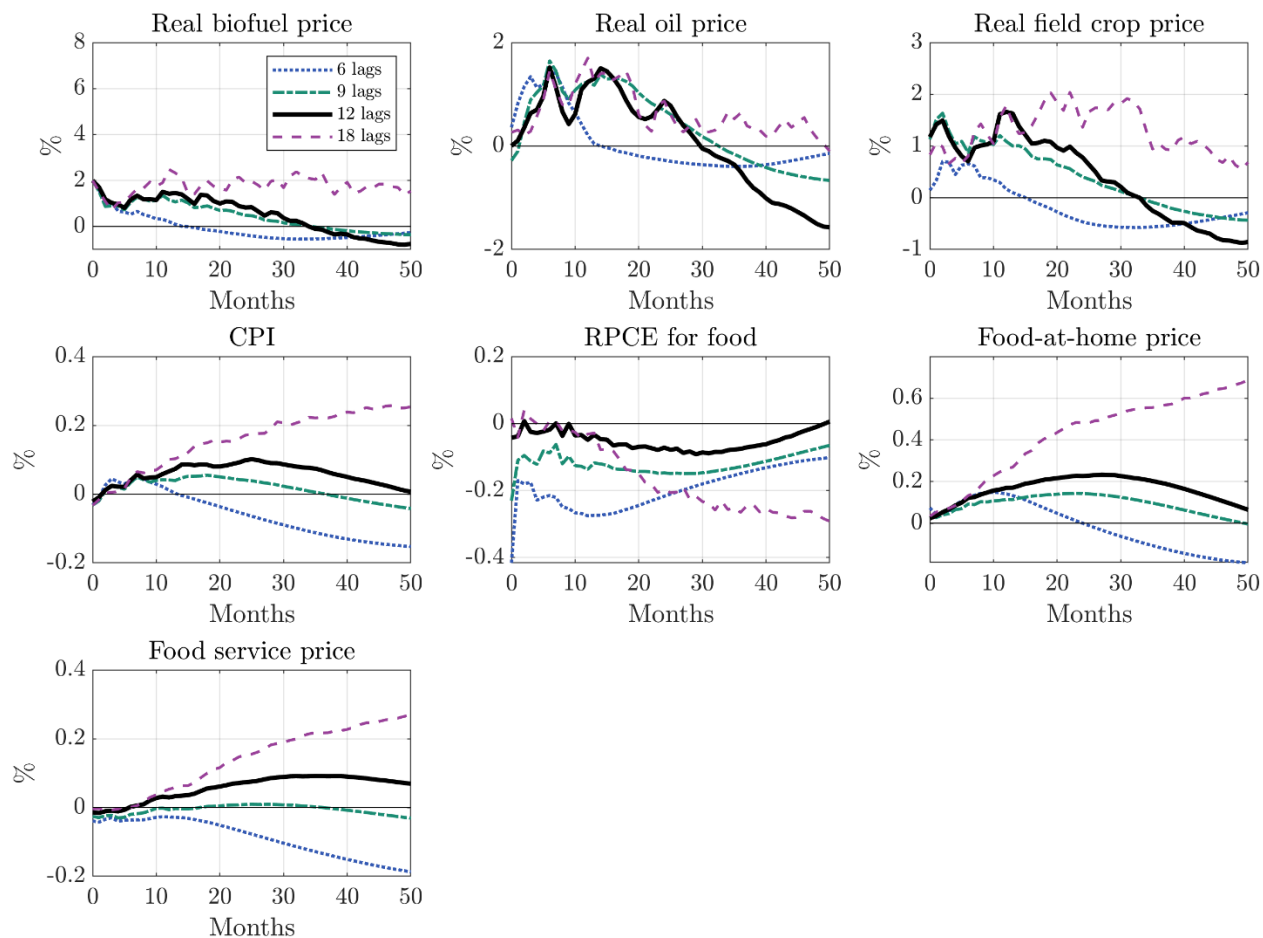
Notes: Impulse responses are estimated using both the local projections and external instruments (SVAR-IV) approach. Responses are normalized to raise the real ethanol price by 2 percent. The green solid line indicates the point estimate from the internal instruments approach, and the black solid line represents the estimate from the external instruments approach. Bands as in Figure 5.

Figure D.3. Impulse responses with ethanol price



Notes: This figure re-estimates the baseline model with the real ethanol price replacing the real biofuel price as the first variable. The shock is normalized to a two percent increase in the real ethanol price (first variable, upper left). Impulse responses are plotted over a fifty-month horizon. The heteroskedasticity-robust first-stage F-statistic is 11.88. Bands as in Figure 5.

Figure D.4. Impulse responses with different lags



Notes: This figure plots impulse responses to a biofuel policy news shock, normalized to raise the real biofuel price by 2%, for three lag orders: $p=3$ (dashed), $p=6$ (dotted), and $p=12$ (solid, baseline). Each line is point estimate; confidence bands are omitted for legibility. First-stage F-statistics are 1.48 and 1.82, and 13.54 for $p=3$, 6, and 12, respectively.

Appendix references

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